Safety, Health & Environment Rules and Regulations

(Mandatory Contract Requirements)

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OBEROI REALTY

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1. Introduction

The Safety, Health & Environmental Protection requirements are business core values of the Employer, and it is applicable to Oberoi Projects and its subsidiary companies here in after referred as "Employer". These Rules & Regulations are mandatory requirements and abiding on all stakeholders including employees, contractors, sub-contractors, vendors, consultants and visitors. The Employer strongly believe that religiously adherence of these rules by the stakeholders should ensure elimination of hazards, reduction in risk and prevention of incidents.

The Employer Project In-Charge, construction managers and project safety managers of the Employer have been empowered to enforce these rules and regulations, should be strictly monitored through periodic supervision, inspection, and audits. Any breach of rules and regulations shall attract appropriate disciplinary action or penalty or both.

Contract safety management flowchart:



2. Applicability of Rules & Regulations

The prescribed Safety, Health & Environment rules and regulations are mandatory for the contractor based on scope of work, hazard & risk, plant & machineries and manpower.

The Contractors should refer to the contract clauses and make sure that safety compliance is achieved, and the objective evidence is available for verification to the satisfaction of the Employer Project In-Charge and Safety Manager.

The Contractor shall be fully responsible for ensuring safety, health and environmentally safe working condition related to contract package in the Employer project premises including prevention of incidents and injury while performing contractual obligations. Any consequences as a result of Contractor's negligence (including but not limited to, accident, injury or death to its workers, legal proceedings, statutory issues and / or property damage, third party loss/damage etc), shall be dealt by the Contractor by himself and the Employer shall be deemed to be indemnified and protected against any losses (including but not limited to, legal and monitory expenses) whatsoever as a result of the aforesaid negligence. The Employer shall have the authority to recover any losses incurred on behalf of the Contractor because of Contractor's negligence from the outstanding amounts due to the Contractor.

3. General Safety and Health Rules

3.1. The Contractor shall abide by the Employer's Environment, Health & Safety Management Manual including EHS Policy, Golden Safety & Health Rules, Standard Operating Procedures and Guidelines. Ignorance to this and deliberate interference with critical safety system or placing others in danger is not acceptable and shall be dealt seriously.

3.2. Contractor must know their "EHS LEGAL" responsibilities under The Building & Other Construction Workers Act 1996 (BOCW Act) & The Maharashtra State Building & Other Construction Workers Rules 2007 (MBOCW Rules) and other common law applicable as mentioned in subsequent clause- Legal Compliance and Register, for their establishment, employees & neighbouring community/ property and shall conduct their business confirming to the best engineering and administrative practices.

3.3. Safety induction and medical examination of new workers and staff is mandatory and shall submit record to the Employer.

3.4. The inspection of tools, plant & machines for its fitness is must. The contractor shall time-to-time notify the Employer Project In-Charge & Safety Manager of the Employer about new deployment of manpower and machines.

3.5. During safety induction, trade skill of workers and operators should also be verified and mention on induction form and ID card.

3.6. The contractor shall provide & maintain all safety requirements prescribed in this document. All cost related to these requirements as directed by the Employer but not limited to the items listed, shall be entirely borne by the Contractor till completion of the scope of work/project.

3.7. The Contractor shall ensure that all stakeholders including worker, staff, sub-contractors, vendors, suppliers, visitors at site has been provided with mandatory safety PPE (safety helmet, safety shoes, safety reflective jacket, Safety harness, Safety goggles, earmuffs) and other necessary PPE as per nature of work & associated hazards and ensure that they shall use it all the time.

3.8. Site access is subject to strict check by the Oberoi Security for PPE, ID Card & legal compliance, the Contractor shall issue photo ID Card by authorized signatory of Contractor.

3.9. Contractor shall perform work as per approved method statement and hazard identification & risk assessment (HIRA).

3.10. Consumption of or working under influence of any alcoholic liquor, illegal drug or other intoxicating substance is prohibited, any violation of rule shall attract eviction of such person from project premises.

3.11. Possession or use of lighter, matchboxes, cigarettes, firearms, ammunition, fireworks, flammable chemicals etc prohibited, other than the one permitted for site work.

3.12. Smoking and tobacco chewing & spitting in the site premises and labour camp is prohibited, any violation of rule shall attract penalty and/or eviction of such person from project premises.

3.13. Fighting, bullying, harassment, horseplay or other violent actions or self-harm or attempted suicide prohibited.

3.14. Ignorance to instructions, rules and signage's and deliberately contravene of safety rules or interfere with critical safety system or placing another employee, contractor or visitor in danger shall be dealt seriously.

3.15. In the event of any emergency or fire broke-out, all personnel should report at nearest assembly point or safe location.

3.16. All height work (above 2m) shall be performed with provision of required safe means of access, safe working platform, safety barricades, lifeline & fall arrestor, safety harness and safety net.

3.17. All Electric Distribution Boards (DB) fixed with 30mA Residual Current Circuit Breaker (RCCB) only should be used. It should have electric danger signage, insulation rubber mat, diagonal two-point earthing to MS panel body and industrial type sockets.

3.18. Contractors are fully responsible to supervise their job, machines and workers and must deploy competent supervisors.

3.19. Use of Bamboo scaffolding is prohibited at The Employer Projects.

3.20. Maintaining good housekeeping at site is basic responsibility of the Contractor. Debris and other construction waste/scrap shall be removed on daily basis to designated location to ensure hygienic, obstruction free and safe working condition at site.

3.21. Dropping of debris and other objects from height/floor or in void or shafts is prohibited and punishable by imposing heavy penalty.

3.22. Contractor shall ensure that all hand tools are secured by tying lanyard to prevent accidental drop from height.

3.23. Shall keep First Aid Box as per BOCW Requirement at workplace and office and display "First Aid" signage prominently.

3.24. Contractor shall submit valid Third-Party Inspection (TPI) Certificate from Competent Person for all lifting tools, tackles, slings, cranes etc., before using these equipment's.

3.25. Contractor shall not remove existing fall protection barricades & safety nets without permission from the Employer safety manager and without arranging alternate protection.

3.26. Hot work permit for gas cutting, welding, brazing etc job is mandatory. Ensure trained workers, basic PPE, welders face shield, ABC type 6kg portable fire extinguisher, fire blanket & flash back-arrester on gas cutter-set. Shall handle and transfer gas cylinders using wheel mounted trolley only.

3.27. Contractor employees shall use urinal, toilet and washing facility. Any violation of urinating /defecating at site premises is serious offence and punishable.

3.28. In the event of Contractor fail to maintain above safety, health and environment requirements, The Employer reserve the rights to ensure or restore safety conditions and back charge the Contractor for its time & cost and administrative charges.

4. Method Statement, Hazard Identification & Risk Assessment and Safety Audit

4.1. Method Statement and HIRA: Contractor shall actively identify the job specific hazards to eliminate or reduce the risk to no harm level. The contractor shall submit method statement after incorporating identified risk and mitigation measures. The hazard identification & risk assessment (HIRA) must be approved by The Employer Safety Manager, Tower In-charge and the Employer Project In-Charge. The contractor's work must proceed according to approved work method statement and any deliberate deviation shall not be acceptable and could lead to stoppage of work and monitory penalty.

4.2. Safety Inspections: Periodic scheduled safety inspection of the workplace and applicable activity viz. construction activity, demolition activity, fabrication activity, waterproofing activity, plant & machines, hand tools, lifting equipment, power tools, electric system, Rope Suspended Platform (RSP) or gondola, hoists, safety barricades, storages, safety nets, PPE's, Welfare facilities, Firefighting system, Batching plant, steel yard etc shall be responsibility of the Contractor. The Contractor shall submit the monthly inspection schedule to The Employer Safety Manager.

4.3. Safety Audit: The civil contractor shall mandatorily conduct third part safety audit through external agency on six monthly frequency and submit audit report and corrective action plan to The Employer. The proposed audit agency should be approved by the Employer safety manager. The contractor shall furnish audit closure report within three weeks after the audit.

4.4. Control Blasting (The Explosives Rules, 2008) for Demolition of existing structure: In case the process involves explosive blasting using control methodology etc., the contactor shall submit appropriate methodology, applicable permissions and compliances as per IS:4081- 1986 safety code for blasting and related drilling operations.

5. Safety Staff

5.1. The contractor is fully responsible for safety supervision of their construction activities and machineries therefore deployment of competent safety staff is must. The contractor shall appoint sufficient number of safety officers as mentioned below but not limited to & approved by the Employer Project In-Charge. The Contractor shall employ only qualified and competent safety staffs as prescribed under BOCW Rules and by The Employer.

S.N.	**Safety Staff**	**Criteria**
i.	DGM/ Sr. Safety Manager (DGM in case of more than 3 Towers)	One DGM/ Sr. Safety Manager as project safety in-charge.
ii.	Safety officer	One Safety Officer per Tower & one for non-tower area.
iii.	Safety Supervisor	One Safety Supervisor per 50 employees or one supervisor per 30 floors (or as per site condition decided by the Employer Project In-Charge)

5.2. Safety Staff Competency Criteria: The Contractor shall employ only qualified, trained and experienced safety staffs as per below criteria:

S.N.	**Safety Staff Category**	**Qualification**	**Experience**
i.	DGM / Sr. Safety Manager	Recognised diploma or degree in engineering or technology or architecture any branch of engineering or BSc. And, Advance Diploma in Industrial Safety (one-year course from State Board of Technical Exam, recognized under State Factories Rules / BOCW Rules or Post Graduate Diploma in Construction Safety Management from NICMAR.	For DGM: Minimum 10 to 15 years' experience in construction safety sector. For Sr. Safety Manager: Minimum 8 to 10 years' experience in construction safety sector. Duties: as laid down in Schedule VI of MBOCW Rules.
ii.	Safety officer	Recognised diploma or degree in engineering or technology or architecture any branch of engineering or BSc. And, Advance Diploma in Industrial Safety (one-year course from State Board of Technical Exam, recognized under State Factories Rules / BOCW Rules or Post Graduate Diploma in Construction Safety Management from NICMAR.	Minimum 5 to 8 years' experience in construction safety sector. Duties: as laid down in Schedule VI of MBOCW Rules.
iii.	Safety Supervisor	Graduate in any discipline and shall possess short term safety & fire management course from NIFE, NILEM, IOSH, NEBOSH etc.	Minimum 3 to 5 years of site safety experience and adequate knowledge of subject.

6. Personal Protective Equipment

It is mandatory at The Employer Projects to wear Safety Helmet, Safety Shoes, Reflective Jackets and where required full body safety harness with double lanyard for height work. The Contractor shall be responsible to provide necessary PPE to their staff, vendor and workers at his own cost and must adhere to Rule 70, 72, 73, 81, 205 etc of MBOCW Rules. All PPEs shall meet quality requirement of applicable Indian Standards and capable to provide complete safety & health protection. The quality shall be checked and approved by The Employer Safety Manager. PPE shall conform to relevant IS Standard / EN Standard as detailed below.

	PPE Items	**Standard and Specifications**
i.	Safety Helmet	MOC: HDPE; IS:2925-1984 (Reaffirmed 2000) or EN Standard 397; chin strap & nap strap (adjustable); Inside- plastic head band. Sticker with name of contractor. Helmet Colour Code: Project staff- White, Safety staff- Green, Workers- Yellow, Operators, Mukadam & supervisor- Blue, Electrician-Red, Visitor- White with "Visitor" sticker, Security- Grey.
ii.	Safety shoes (with steel toecap)	IS:15298-2002; EN20 345 certified by DGMS/CLI Lab; ankle height not less than 7cm, acid/alkali resistant sole; foam cushion in inner side for ankle lap.
iii.	Reflective safety jacket with fluorescent strips	Staff- Green and Workers- Orange colour of suitable make and washable with Contractor's name/logo printed on it.
iv.	Full body industrial safety harness	IS:3521-1999/EN 361; Standard lanyard 1.8m; scaffold-hook and double lanyard
v.	Fall arrestor & Lifeline	Fall arrester: EN 353-2:2002 Lifeline: 12mm/14mm polyamide rope
vi.	Earmuff / Ear Plug	IS: 6996-1973 (Reaffirmed 1998); IS 9167 or EN 352-1:2002 and EN 352-2:2002; made of sponge or foam; preferably with cord; length of cord not less than 50cm; after squeezing it should return to its original shape within 30 seconds.
vii.	Safety Goggles	IS: 5983-1980 (Reaffirmed 2002) or EN 166:2001; made of tough polycarbonate material; lenses with anti-scratch treatment; colour of lenses- Clear (UV clear).
viii.	Face shield	IS: 8521 (Part II) - 1977 (Reaffirmed 2002) IS: 8521 (Part I) –1994 (Reaffirmed 2002) or EN 175F; IS-1179:1967.
ix.	Respirators/ Dust mask	IS: 15321- 2003, IS: 15322 - 2003 and dust mask IS:9623
x.	Hand Gloves (canvas)	IS 6994-Part-I; MOC-split or chrome leather; stitching should be firm; Cut-resistance proof
xi.	Hand Gloves (leather)	IS6994 Part-I; MOC- Coated fabric; additional lining at palm; firm stitching. Cut-resistance proof
xii.	Electricians hand gloves (rated for voltage)	IS-4770:1991-rubber gloves

	PPE Items	Standard and Specifications
xiii.	Shoulder pads for reinforcement handling	Round neck type leather covered with foam cushion, as per manufacturer's specifications.
xiv.	Gum Boots (with steel toecap)	IS-5557: 2004; IS 12254: 1993 (PVC)
xv.	Electricians' safety shoes (with non-conductive toecap)	As per manufacturers specifications and test certificate

6.1. Contractor shall always maintain minimum 25% buffer stock of PPE with respect to total manpower at site and stock status shall be displayed. In case contractor fail to provide PPE, The Employer reserve right to stop the work and impose penalty.

6.2. Contractor shall handover PPE to staff and workers immediately after safety induction and shall demonstrate correct usage of PPE & safety harness.

6.3. It is mandatory that all supervisors, site engineers and visiting vendors should wear safety helmet, safety shoes and reflective jackets and use safety harness for all height work as required.

6.4. Workers employed on mixing asphaltic materials, concrete, cement and mortars shall use rubber gum boots with toecap & hand gloves and other PPE like goggles, respirators and hand gloves, etc.

6.5. Persons engaged in welding and gas cutting work shall use appropriate welding face shields, leather hand gloves and fire-retardant protective cloths. Welding/Gas cutting helper shall also provide and use appropriate eye protection goggles.

6.6. Workers engaged in rock breaking, grinding, drilling and chipping etc shall use protective eye goggles, dust respirators, dust mask, earmuffs /ear plugs, etc.

6.7. Persons working at height and exposed to risk of fall from height shall use full body safety harness with double lanyard & kinetic shock absorbers, fall grab-arrestor and lifelines ropes as per the risk.

7. Fire Prevention and Protection

The contractor shall take all necessary precautions to prevent outbreak of fires at the construction site and at the same time keep adequate provisions to extinguish the fire. Any fire risk and personal or property damages posed out of negligence or ignorance to rules by contractor shall be at cost and risk of respective contractor. Fire is most successfully fought when the extinguishing equipment is suitable, and persons are trained. Following rules shall be adhered:

7.1. Fire extinguishers suitable for the different class of fires such as Class A, B, C & D as per IS: 2190: 2010- Selection installation & maintenance of first aid fire extinguisher and sand buckets shall be made available at designated "Fire Points" at each floor and in podium & basements at every 15m with unobstructed access marked by location signage. The contractors fire protection layout shall be approved by the Employer Safety Manager.

7.2. Impart training to all employees for efficient firefighting with hands-on training and six-monthly refresher training.

7.3. The major Contractor shall furnish list of trained people in firefighting to the Employer and display list at site and office.

7.4. The contractor shall adhere to MBOCW - Rule 62: Fire Protection and shall make all necessary provision for prevention & fire fighting.

7.5. As per The Employer Rules entire construction site is “No Smoking Zone” and should display warning instructions and signages prominently at conspicuous places. Any ignorance shall be treated as major violation and person/contractor will be barred from entering site. The Contractor shall communicate fire safety rules to staff & workers during safety induction.

7.6. All hot work at site like welding, gas cutting, brazing, heating etc shall be undertaken under Hot Work Permit & under supervision of contractor’s safety officer.

7.7. Quantities of combustible/ flammable materials like timber, shuttering ply, paints, diesel, gas cylinders, cardboard boxes etc. shall be the minimum required to reduce the fire load to minimum.

7.8. Containers of flammable paints, thinners, diesel, grease and allied materials shall be stored in a separate room at ground level which shall be well ventilated and free from excessive heat, sparks, flame, any type of hot work or direct sunrays and all containers of volatile liquids should be always kept closed by lids. The store shall be clearly marked as “Flammable Material”, and Foam Type Extinguishers and Sand Buckets shall be located near such storage.

7.9. The Portable Fire Extinguishers, Fire Buckets, Fire Hose-Reels, Fire Water Tank, Fire Pump, etc shall be inspected once in a month for fitness conformity and logbook shall be maintained. The date of last inspection and maintenance shall be displayed prominently on fire extinguisher body.

7.10. All pressurised fire extinguishers shall have valid hydro pressure test certificates confirming IS 2190: 2010-Selection installation & maintenance of first aid fire extinguisher.

7.11. The Contractor shall conduct fire emergency drills and site evacuation on quarterly basis.

7.12. The list of emergency telephone numbers including fire brigade number shall be prominently displayed at suitable locations (such as site office, meetings rooms, canteen, tower entrance lobby and staircases, gate security rooms, passenger hoist, first aid centre, stores etc.) in bold distinct font.

7.13. All hot work shall be backed by 6 kg DCP/ABC Type portable extinguisher.

7.14. Fire-fighting equipment should be used for fire-fighting purposes only, never place obstructions within 15 ft. of fire extinguishers and hose boxes.

7.15. The Civil Work Contractor shall install suitable firefighting system including active hose-reels installed at every alternate floor supported by adequate back-up of fire water supply, pressure pump and alternate electric supply. The scheme should be approved by MEP Head and Project In-Charge of the Employer.

8. Housekeeping and Hygiene

8.1. The Employer value Good Housekeeping as foundation stone for good safety culture and management believe that many incidents can be avoided just by ensuring clean and orderly construction site. The contractor must adhere to 5S Culture and Housekeeping Guidelines.

8.2. The respective Contractor shall be responsible for housekeeping and proper material stacking etc of their working place and surrounding area and should not be obstruction & unsafe for others.

8.3. Housekeeping management shall be part of the Site EHS Plan and Site Logistics Plan and should construct appropriate size of waste collection yard with signages for steel scrap, wooden scrap, plastic, glass & construction debris etc.

8.4. The contractor shall stack all undesirable construction waste/ debris at designated scrap yard at his own cost, as directed by the Employer.

8.5. The contractor shall follow the recommendations of IS: 4082-1996 (Reaffirmed in 2003) for stacking and storage of construction materials and components at site.

8.6. The planning and construction of temporary structures shall be such that they do not hamper easy movement of worker, vehicles & cranes. No materials on any of the sites of work shall be so stacked or placed as to cause inconvenience to any person or the public.

8.7. The Employer Project In-Charge and Safety Manager have authority to stop work if Contractor/s fails to improve housekeeping after having been notified.

8.8. The Employer may instruct the contractor to remove any materials which are of danger or cause inconvenience to the public.

8.9. After completion of the assigned work, the contractor shall remove all scaffoldings, surplus materials, scrap, rubbish and all temporary structures, huts and sanitary arrangements used/installed for his workmen at site.

8.10. Contractor shall deploy sufficient housekeeping manpower as per the size of the project but not less than 5% of total workforce and should be identified suitably by different colour reflective jackets as "housekeeping Team" (recommended "yellow" colour jacket with reflective bands), supervised by Responsible Person of the Contractor. The Responsible Person should schedule housekeeping of construction area, perimeter, internal access roads and all associated places such as offices, stores, workshops, stock yards, batching plant, labour camp etc and maintain a register.

8.11. The major Contractor shall ensure separate gates & access roads for labour entry and material vehicles to prevent workers directly encountering vehicle traffic.

8.12. The walkways, gangways shall be ensured with minimum width of 1.2m and shall be clear of debris, protruding dowels, timber with nails, oil spillages & other obstructions all the time and prevent tripping danger. Similarly, all emergency exit, fire doors, firefighting equipment, access to electrical panels & DB's, first aid stations, ambulance station, stairways, ladders, scaffolds etc should be unobstructed, dry and in good working order.

8.13. The Contractor shall ensure adequate illumination in the workplace and maintain general uniform illumination of minimum 50 Lux across length & breadth including staircases, passages etc.

8.14. The contractor shall install a construction vehicle tyre washing facility and vehicles leaving the site should be cleaned by water jets, specially wherever there is a possibility of spillages on public roads, particularly during excavation, piling, rainy season etc. Also, contractor should ensure cleaning of public roads in case of spillages of soil, muck, material etc. (Applicable only to 'Package A' and 'Package D' Contractor).

8.15. The Contractor shall ensure that parking of trucks, transit mixers& other construction vehicles not stranded on public roads, which may impair aesthetics & obstruct traffic.

8.16. Dewatering of stagnant water from site shall be regular activity equipped with adequate capacity of pumps.

8.17. It shall be daily routine that workplace is cleared off all waste / debris and disposed into waste bins after the work & before leaving the workplace.

8.18. The contractor shall on weekly basis or as required remove construction waste, debris, scraps, waste oils, packing waste, empty drums, etc, from working place/ upper floors to waste collection yard or more frequently depending upon the quantum or as directed by the Employer Project In-Charge.

8.19. Contractor shall arrange to clean weekly - safety nets, catch platforms, chajjas, link-beams, etc. and strictly ensure it is free of debris & other material.

8.20. Similarly, all surplus earth and debris shall be removed from the project premises to legally permitted dump yard or as per logistics plan.

8.21. Dropping of debris from floors or any height is strictly prohibited and punishable act. The contractors should use passenger hoist or staircase for transporting waste from upper floors safely, dully packed in the gunny bags.

8.22. Over and above as a good practice the Contractors must plan at least a day fortnightly as "Housekeeping Day" to keep the site orderly.

8.23. Electrical cables, welding cables shall be routed overhead to prevent trip/fall hazard & electric shock. Overhead cable routing height should be minimum 2.5m inside the buildings and minimum 6.1m in open area supported by stable & insulated stands/MS-poles or ensure that DB's and welding machines are placed near by job location.

8.24. Materials shall be stacked at least 2m away from leading edge of slabs and floor openings (MEP openings, lift shaft openings, roof edges, excavations, or trenches, scaffolding platforms etc.) to prevent fall of material to below level.

8.25. The stock yard shall be planned as part of logistic plan with proper layout ensuring safe access for receiving / shifting of material by vehicles, cranes and stacked in an orderly manner & segregated.

8.26. Steel reinforcement shall be stacked in the MS stands/racks within the safe limit.

8.27. Marble, granite etc., material shall be stacked in A-Frame within storage limit and secured with ropes/ratchet strap belt to prevent toppling of marble stacks.

8.28. Steel, shuttering & other materials etc shall not be stacked more than 1.5m height and concrete block and bricks should be stored to a height not exceeding 1.8m at ground level; however, while stacking material at upper slab level consider the load bearing capacity of structure.

8.29. Piles stack of cement bag shall not be more than 10 bags in height unless such stack piles is stacked in suitable enclosure or otherwise adequately supported.

8.30. Sites having space constraints shall limit the material inventory or staggered inventory, disposed of scrap/ debris & other excess or waste material more frequently and plan rack stack methods.

8.31. Ensure that the work vicinity of hot work is cleared of all combustibles and rubbish and hot work permit procedure is strictly adhered.

8.32. Signages: The contractor shall display required nomenclatures and mandatory, cautionary & informative signage such as speed limit, no smoking, electricity danger sign, flammability diamond sign, no spitting, drinking water, urinal/ toilet with gender pictorial sign, empty & full gas cylinders storage, emergency exit, assembly point, numbering of floors/ levels etc as well as display caution signages for height work, falling objects, deep excavation, fire, electrocution, unauthorized entry or as per site requirement etc prominently.

8.33. The contractor shall employ full time sufficient sweepers to maintain latrine and urinal blocks in clean and hygiene condition and maintain logbook.

8.34. It shall be cleaned and disinfected at least every day and maintained properly by the contractor throughout the project duration.

8.35. Activities inside the confined space or in any enclosed environment like surface grinding, buffing and other similar dust emission work shall be supported with mechanical dust extraction system appropriately as per the work requirement and approved by the employer project in charge.

8.36. Housekeeping management shall be part of the Site EHS Plan and Site Logistics Plan and should construct appropriate size of waste collection yard with signages for steel scrap, wooden scrap, plastic, glass & construction debris etc.

8.37. The major Contractor shall ensure separate gates & access roads for labour entry and material vehicles to prevent workers directly encountering vehicle traffic.

9. Lifting and Hoisting Machinery

The use of lifting machines and tools/tackles including attachments, anchorage and supports shall conform to the Rules 82 to 108 of chapter XI of MBOCW Rules, 2007 and shall also conform to the following conditions.

9.1. Lifting machines and tackles shall be of good mechanical construction, sound material and adequate strength and free from any defects and shall be kept in good repair and in good working condition. Every sling used in hoisting or lowering materials or as the means of suspension shall be as per manufacturer's guidelines, of good quality and adequate strength and dimension and free from any defect. Test certificates of such ropes, slings, hook, D-shackles, etc. shall be submitted in advance to the Employer Project In-Charge.

9.2. The crane operator and other lifting appliance operator shall be properly trained and competent. No person under the age of 21 years shall be in charge of any hoisting machine or to give signal to operator of such machine.

9.3. In the case of every lifting machine (and of every chain, ring, hook, D-shackle, swivel, and pulley block used in hoisting or as means of suspension) the safe working load (SWL) shall be ascertained and clearly marked. In case of a lifting machine, having a variable safe working load, each safe working load and the conditions under which it is applicable shall be clearly indicated. No part of any machine or any gear shall be loaded beyond the safe working load except for the purpose of testing. This shall be approved by the P&M In-charge and Safety Manager of the contractor.

9.4. The base of such hoisting equipment shall be kept in perfect horizontal condition since any tilt would reduce the load carrying capacity of the equipment. The foundation shall be firm enough to support the equipment. The level shall be checked every day before starting the work in case of mobile hoisting equipment.

9.5. Double sling and four leg slings shall be used for hoisting material. The angle of the sling shall be wide enough for safe hoisting and the sling shall be adjusted as per the centre of gravity of the material to be lifted. A guide rope (manila rope of sufficient length, normally 1.5m long) shall be attached to the end of the material lifted to pull the same conveniently during lowering.

9.6. The Contractor shall notify the safe working load (SWL) of the lifting machine to the Employer Project In-Charge whenever new machinery brought at site and submit TPI certificate by Competent Person, without which machines shall not be taken into operations.

9.7. The contractor shall arrange thorough inspection and load testing of all lifting machines and tackles by the Competent Person, at least once in every six months and the records of such inspection and testing and updated inventory shall be maintained, and a copy shall be submitted to The Employer.

9.8. Motors, transmission, couplings, belts, chain drives and other moving parts of hoisting appliances shall be provided with adequate safeguards.

9.9. The contractor shall maintain a Register of Periodical Third Party Test (TPI) and maintenance history card of all Lifting machines, hoists and Gears at site as per Rule 97, Chapter XI, MBOCW Rules and submit the same to the Employer Project In-Charge periodically for examination & record.

9.10. The contractor should submit the preventive maintenance schedule of Plant and Machineries and deploy adequate competent staff for periodical inspection, maintenance and servicing. Ensure annual servicing of the plant & machineries from original equipment manufacturer (OEM).

9.11. All vehicles and earth moving equipment are made of good material, proper design and sound construction and are sufficiently strong for the purpose for which such equipment are used and are maintained in good state of repair and are properly used in accordance with standard safe operating practices

9.12. A shovel or an excavator used for excavation work is constructed, installed, operated, tested and examined as required under any law for the time being in force and the relevant national standards.

9.13. During the excavation, excavator swing zone is restricted for any movement, controlled by a dedicated signaller/helper.

9.14. An operator of a bulldozer before leaving such bulldozer shall applies the brakes, lowers the blade and sipper and shall puts the shift lever into neutral.

9.15. A bulldozer is left on level ground at the close of the work for which such bulldozer is used. The blade of a bulldozer is kept low when such bulldozer is moving uphill. The bulldozer blades are not used as brakes except in an emergency.

9.16. The truck or trailer employed for transporting freight containers are of the size sufficient to carry the containers, without overhanging and are provided with twist locks conforming to national standards, at all the four corners of each of such truck or trailers and such truck or trailers are certified for such use by an authority under the relevant law for the time being in force and is inspected by a responsible person, at least once in a month and record of such inspection is maintained

9.17. All transport or earth moving equipment and vehicles are inspected at least once a week by a responsible person and in case any defect is noticed in such equipment or vehicle, it is immediately taken out of use.

9.18. Power tracks and tractors are equipped with effective brakes, head lights and tail lamps and are maintained in good repair and working order.

9.19. Side stanchions on power trucks and trailers for carrying heavy and long objects are - (i) of sound construction and free from defects; (ii) provided with tie chains attached to the top across the loads for preventing such stanchions from spreading out; and (iii)kept in position while loading and unloading.

9.20. Safe gangways are provided for to and fro movement of building workers engaged in loading and unloading of lorries, trucks, trailers and wagons.

9.21. Trucks and other equipment are not loaded beyond their safe carrying capacity which shall be clearly marked on such trucks and other equipment.

9.22. Handles of hand trucks are so designed as to protect the hands of the building workers working on such trucks, or such handles are provided with knuckle guards.

9.23. No unauthorised person rides the transport equipment employed in such work.

9.24. A driver of transport equipment manoeuvres such equipment under the direction of a signaller/helper.

9.25. Adequate precaution such as isolating the electric supply or erecting overhead barriers of a safe height is taken when earth moving equipment or vehicles are required to operate in dangerous proximity to any live electric conductor.

9.26. Vehicles and earth moving equipment's are not left on a slope with the engine of such vehicles or equipment running. A wheel chock is provided while parking the vehicles.

9.27. All earth moving equipment's, vehicles or other transport equipment are operated only by such person who are adequately trained and possessed skill and valid driving license as are required for safe operation of such equipment, vehicle or other transport equipment.

9.28. Every vehicle or earth moving equipment is equipped with - (i) silencers (ii) taillights (iii) power and hand brakes (iv) reversing alarm and (v) search light for forward and backward movement, which are required for safe operation of such vehicle or earth moving equipment.

9.29. Provide one portable ABC type fire extinguisher (6kg) in every earth moving machines and dumper vehicles.

10. Compressed Gas Cylinders

10.1. The Gas Cylinders together with their valves and other fittings and identification colours shall be maintained in good condition.

10.2. No cylinder shall be subjected to any heat treatment or exposed to a high temperature or under direct sunlight or stored with flammable or explosive material.

10.3. Every cylinder containing compressed gas shall have its valve securely closed to prevent leakage. Valves fitted to the cylinders containing LPG and highly toxic gases shall be provided with security nut on the outlet to act as a secondary means of safeguard against leakage of gas.

10.4. If the leak in the valve cannot be rectified by tightening the gland-nut or the spindle, the cylinder shall be removed to an open space where it is least dangerous to life and property and the Supplier shall be informed.

10.5. Domestic LPG gas cylinders are not allowed to use at the Employer site.

10.6. Cylinders shall be stored in upright position and adequately secured.

10.7. Gas cylinder handling shall be done by using wheel mounted trolleys or cradles and cylinders should not be dragged or rolled. The cylinder set should remain on trolly while at workplace.

10.8. The cylinders shall be handled carefully and not be allowed to fall on one another or subjected to any undue shock.

10.9. Sliding, dropping, rolling or playing with cylinders is prohibited.

10.10. Cylinders shall be stored in cool, dry, well-ventilated place under cover, away from open flames or any potential sources of heat
10.11. The storage room or shed shall be of fire-resistant construction and well ventilated.
10.12. Liquified, thin-walled cylinders such as LPG and dissolved gas (DA) shall not be stacked in horizontal position. All cylinders shall be stored in up-right position only.
10.13. Cylinders containing flammable and toxic gases shall be kept separated from each other by 3m and from cylinders containing other types of gases by an adequate distance or by suitable partition wall.
10.14. Cylinders shall not be stored with any combustible materials.
10.15. Empty cylinders shall be segregated from filled ones with signage and care shall be taken that the valves are tightly shut.
10.16. The contractor shall ensure that hydro-static test certificates of all pressurized gas cylinders are obtained & record available.
10.17. No lifting magnet shall be used in loading or unloading of cylinders filled with compressed gas. When any such operation is carried out by means of a crane or forklift truck, a proper cradle of cylinder height should be used as approved by safety manager.
10.18. In case the storage of number of gas cylinders exceeds the allowable limit, contractor shall obtain the required license from authority.
10.19. The contractor should abide by all the safety precautions mentioned in IS 1323:1982 Code of practice for oxy-acetylene welding for structural work in mild steel.

11. Electric Arc Welding and Gas Cutting (Hot work)

11.1. All the hot work shall be carried out strictly after obtaining Hot Work Permit from the Employer Site Engineer.
11.2. The contractor shall ensure site is cleared off all combustibles and use fire blankets to contain weld spatters and prevent fire.
11.3. For welding sets, all power connections shall be routed through RCCB of 30mA rating DB. Double earthing shall be ensured to the welding machine. A provision of a separate return lead shall be ensured connecting welding object and machine.
11.4. The cable to be used shall be of adequate capacity corresponding to output of the welding transformer / generator and shall be routed through dry isolated path. Welding cable terminals shall be provided with lugs and connected properly. Proper insulation of cable with insulation tape of approved quality shall be ensured and only double insulated cable shall be used. Extension of welding cables shall be done using standard connectors.
11.5. Personal contact with the electrode or other live parts of electric welding equipment shall be avoided. Wires and cables shall not be hung from any metal hook.
11.6. Accidental contact of electrodes with ground shall be prevented.
11.7. The welding cables shall not be allowed to get entangled with power cables. It shall be ensured that the cables are not damaged by movement of materials. Dragging and coiling of cable shall be avoided.
11.8. Barrier screens shall be erected to protect other persons from harmful rays and sparks from the work. When welding or gas cutting is carried out in elevated positions, precautions like providing metal sheet, fire blankets etc. shall be taken to prevent sparks or hot molten metal falling on persons or flammable/ combustible materials below
11.9. Adequate ventilation shall be provided for easy dispersion of toxic fumes while welding, brazing and cutting in confined space.
11.10. Suitable type of protective clothing consisting of fire-resistant gauntlet gloves, boots and aprons shall be provided to workers to protect from heat and hot molten metal splashes. Welding shields with filter glasses of appropriate shade shall be worn as face protection against UV & IR rays.

11.11. Welding and gas cutting shall not be carried out by standing on drums, barrels, tanks or other containers. Welding and gas cutting shall not be carried simultaneously with the painting work.
11.12. Gas cutting set shall be equipped with flash back-arrester and NRV. Gas cutting hose shall be checked for any leakage using Soap bubble test.
11.13. Only industrial Gas Spark Lighter shall be used. No match box or other means will be allowed for igniting.
11.14. During the hot work in MEP shafts of service pipelines, the bottom end of the pipes shall be closed by blind flange to prevent the spreading of spatters and subsequent risk of fire.
11.15. One 6kg ABC type portable extinguisher shall be compulsorily stationed near welding/ gas cutting work.
11.16. Contractor shall ensure that the hot work location and surrounding is inspected by fire watch at least for one hour after the hot work is over.

12. Electrical Safety

12.1. Before commencement of construction work, the Contractor shall take adequate measures to prevent worker coming into physical contact with any electrical equipment or apparatus, machines or live electrical circuit which may cause electrical hazard.
12.2. The Contractor shall display and maintain suitable danger warning signs on all electric panels and DB's in Hindi or local language understood by the majority.
12.3. Where exact location of underground electric power line is not known, the building workers using jack hammers, crow bars and other hand-tools which may come in contact with a live electrical line, shall be provided with insulated tools, protective gloves and footwear.
12.4. At excavation site, the Contractor before using excavators and other excavating equipment shall conduct survey to trace live cables and disconnect them.
12.5. The Contractor shall ensure that, no electric wiring, which may come in contact with water, or which may damage mechanically is left on ground or floor.
12.6. Contractor shall ensure that all temporary electrical installations are provided with 30mA RCCB only, including dewatering pumps.
12.7. Main electrical panel shall be equipped with Earth Leakage Relays (ELR)
12.8. The Contractor shall deploy licensed electrical staff (approved by Government licensing authority) at site and submit list to The Employer.
12.9. The Contractor shall provide required PPE to electricians as per Indian Electricity Rules.
12.10. It shall be the responsibility of the Contractor to provide and maintain complete installation on the load side of the supply point with regard to the safety & maintenance.
12.11. All cabling and installation shall comply with statutory requirements i.e. The Electricity Act, 2003; Indian Electricity Rules, 1956 (as amended in 2005); National Electric Code 1985 (as amended in 2005); Other relevant rules of Local Bodies and Electricity Board.
12.12. The Contractor shall submit schematic and single line diagram for approval of the Employer and display it on Panel/DB with name & number of authorised electricians.
12.13. Method of earthing, installation and earth testing results shall conform to IS Specifications [IS: 3043 - 1987 (Reaffirmed in 2001)].
12.14. Electrical equipment and installations shall be installed and maintained as to prevent danger from contact with live conductors and to prevent fires originating from electrical causes like short circuits, overheating, etc.
12.15. The minimum clearance to be maintained from HT overhead lines shall be 6.10m (minimum) as per the Rules 77-80 of Indian Electricity Rules, 1956 (Amended in 2005).
12.16. No material or earth work shall be allowed to be dumped below or in the vicinity of the bare overhead line conductors, which may hamper minimum safety clearance.

12.17. Grounding conductor of wiring system shall be of copper or other corrosion resistant material. A double grounding connection shall be made in appliances/equipment where chance of electric shock is high.
12.18. Electric fuses and/or circuit breakers installed in equipment circuits for short circuit protection shall be of proper rating and approved by MEP In-charge.
12.19. Wires and cables shall be adequately supported, and an approved method of fixing shall be adopted. Loose hanging wires & cables shall be avoided. Lighting and power circuits shall be kept distinct and separate.
12.20. Reinforcement rods, props, scaffoldings or any construction metallic part of structure, trees' trunk or branches shall not be used as support for wires and cables, fixtures, equipment, earthing, etc.
12.21. Overhead cables shall be routed minimum 6 m height on insulated MS poles/ stands and all cables inside buildings shall be routed 2.5m height to prevent trailing, tripping and electrocution. Floors shall be free of trailing cables to avoid tripping/ electrocution hazard.
12.22. The Site Distribution Boards and extension boards shall be as per IP-55 specification i.e. dust & water ingress proof mounted on stand or wall and should have individual switch for each socket.
12.23. Industrial type three-pin plug only shall be used. Open/naked wire connections are prohibited.
12.24. Open type and wooden frame DBs are prohibited to use at the Employer projects.
12.25. Isolating switches shall be provided close to equipment for easy disconnection of electrical equipment or conductors from the source of supply in emergency and repair including for De-watering Pump.
12.26. All connections to lighting fixtures, starters or other power supplies shall be provided with PVC insulated, PVC sheathed twin/three/four core wires to have better mechanical protection for preventing possible damage to equipment or injury to personnel. Taped joints in the cables shall not be allowed.
12.27. All the lighting fixtures and lamp holders shall be of good quality and in good condition.
12.28. Only PVC insulated and PVC sheathed wires or armoured PVC insulated and sheathed cables shall be used for external temporary power supply connections.
12.29. Lamps used for illumination and testing purpose shall have cover or guard to protect them from accidental breakages. Only 24 Volt supply system shall be used for hand lamps etc, while working inside metallic tanks, vessels or confined space.
12.30. While working on or near a circuit, whenever possible the use of one hand may be practiced even though the circuit is supposed to be dead. The other hand may preferably be kept in pocket.
12.31. Operation of electrical equipment and welding equipment shall be avoided when standing on wet floor or when hands are wet.
12.32. While tapping electricity from the socket, three-pin plug top must be used. It shall be ensured that no extension boards are overloaded while tapping.
12.33. Use of domestic two core cables and PVC extension boards are prohibited.
12.34. Power supply to machines and lighting fixtures shall be switched off when not in use.
12.35. Temporary electrical connections shall be removed as soon as the stipulated work is over. After completion of the works, the contractor shall disconnect /dismantle the DB or extensions and the other facilities he may have erected.
12.36. Electrical Safety work permits & LOTO (lock out- tag out system) shall be used for switching off the main feeder and equipment by the contractor during batching plant maintenance, pan-mixer cleaning, tower crane & other machines maintenance.
12.37. Power tapping for electrical hand tools shall as near as possible to prevent trailing cables.
12.38. Inspection & testing: RCCB shall be tested monthly, earth pits & earthing connections shall be tested quarterly basis and record shall be evident.
12.39. All earth terminals shall be visible and earth pits shall be identified and numbered. Neutral conductor shall not be treated as earth wire.
12.40. All three-phase equipment's, MS cabinets and Machine MS body shall be double earthed diagonally or as indicated by the machine supplier.

12.41. All light fixtures and portable equipment shall be effectively earthed to main earthing.
12.42. The entire installation shall be subjected to the following tests before energizing of installation including portable equipment: Insulation resistance test, Polarity test of switches, Earth continuity test, Earth electrode resistance.
12.43. In front of DB's a clear space of 1.0 m shall be maintained as emergency access.
12.44. All electrical equipment to be used in explosive environment shall be explosion proof.
12.45. As per provision of Rule 47-A of IE Rule 1956, license to operate the DG shall be mandate to contractor.
12.46. First Aid treatment chart of electrical shock shall be displayed at First Aid Centre, Panel room, substation etc.

13. Portable Electrical Hand Tools & Equipment's

13.1. Portable electrical equipment shall be weekly examined, tested and maintained to ensure that the equipment and its leads are in good order. Log sheet shall be maintained for inspection recording the inspection dates & faults noted. The insulation and winding resistance of the portable electrical equipment shall be checked at least once in a month and report shall be submitted for all such machines.
13.2. All portable appliances shall be provided with dead man switch, three core double insulated cables and three pin industrial plugs. It shall be ensured that the metal part of the equipment shall be effectively earthed.
13.3. Connections to portable equipment or machines from the panel/distribution board/extension board shall be taken using 3 core double insulated PVC flexible copper wire in one length. No joints shall be allowed in this flexible wire. In case, single length of wire is not sufficient for a location then the supply can be tapped by providing another extension board comprising of switch and socket.

14. Hand Tools Safety

14.1. Impact Tools: The contractor shall use precision grip for the most commonly used impact tool, hammer for light work. For safe operations, the hammer shall have a straight cylindrical handle of 24 to 40 mm calibre with a maximum length of 600 mm and maximum head weight of 6.5 to 7.5 kg. Hammers shall be maintained such that cracked or weak handles are replaced, and heads are in good condition and firmly secured to an undamaged handle.
14.2. Cutting Tools: The contractor shall ensure that various cutting tools like axes, chisels and shovels, etc. are made up of material with adequate strength. The contractor shall ensure that wooden handles are to be moist before use during summer. Proper PPEs like hand gloves, ear plugs, goggles, dust respirators, etc. shall be provided to the worker as per the need of the work.
14.3. All hand tools shall be secured by tying lanyard to prevent accidental drop from height.

15. Machine and Hand Tools Guarding

15.1. The Contractor shall ensure that rotating, reciprocating & dangerous parts of machines and hand tools securely guarded, and they are never removed while machine is in motion and inspected weekly by Contractor.
15.2. On completion of maintenance, oiling/ greasing any guards that was removed must be replaced, before machine again taken into operation.
15.3. Construction of guards shall be as per IS 9474, which shall prevent human body access to the rotating parts of machineries.
15.4. Grinding wheels of prescribed diameter and IS quality only shall be used, in order not to exceed the prescribed peripheral speed.
15.5. Safety goggles shall be worn during grinding/cutting operations. Adhere Rule 64 MBOCW Rules.

16. Fall Prevention and Protection

Fall prevention measures is the foremost priority of the Employer. The Contractor shall make sure all efforts to achieve this objective and ensure robust proactive and positive measures. The use of full body safety harness is minimum necessity; however, it is the last level of protection and is not the substitute to the hard barricades, safety gates, leading edge safety railings, safe working platforms, toe-guards, access staircase, ramps, safety nets and lifelines etc.

As a Mandatory Rule, the Contractor shall not remove any physical safety protection measure without the permission of The Employer Safety Manager and any deviation which may endanger life of people, shall attract stoppage of work & heavy penalty.

16.1. Barricades and Railing Standard

Basis building floor designs the contractor shall identify and prepare fall protection plan and get it approved from the Employer Project In-Charge. The excel tracker of fall protection plan prepared in tandem with the Construction Planning Schedule, of horizontal & vertical openings such as MEP cut-outs, other services shafts, staircases, voids, lift shafts, windows, slab edges would be handy plan for Contractor for fabrication schedule. The protection fixtures shall be ready before shuttering work of the floor is initiated. The contractor shall ensure that railings & barricades shall be of following specifications and must meet BOCW & IS 4912 (1978): safety requirements for floor and wall openings, railings and toe boards, requirements.

S. N.	Fall Protection Requirements	Specifications & MOC for Safety Barricades and Railings
i.	Leading edges of constructed floors, top working floor leading edges, staircases; voids, MEP cut-outs, services shaft cut-outs, lift shafts, open balconies, temporary platforms, scaffold railings, suspended platforms, elevated platforms, projected decks etc, connecting walk-way beams, refuge floors slab edges etc.	**MS Pipe Railings:** **MOC: Minimum 40NB MS pipes, scaffold clamps & baseplates, fasteners and vertical monofilament/ PVC net cover**: a) Top horizontal guardrail at 1150mm height b) Intermediate guardrail at 550mm c) Vertical posts at not more than 3m apart. d) Vertical Posts: Base plate 150 x 150 mm, rigidly fixed on floor by anchor fasteners (10 mm). e) Connector for railing pipes: scaffold clamps/ nut-bolts/ welding. f) Toe-board of 150mm height. g) Strength of railings: Capable to bear lateral load of about 100kg. h) Neatly painted in red & white bands. i) Periodical inspection & maintenance for corrosion, welding defects, damages and replacement, repainting etc. **OR** **Wire Rope Railings: for opening width less than 5 Meters (Such as Window, Ducts, Service shaft & doors).** **MOC- Three-layer rope fall protection** a) Wire rope 10mm MSGI, b) Top layer at 1200 mm

S. N.	Fall Protection Requirements	Specifications & MOC for Safety Barricades and Railings
		c) Mid layer at 700 mm d) Bottom layer at 200 mm from surface/slab e) Toe-board of 150mm height. f) Strength of railings: Capable to bear lateral load of about 100kg. g) Rated Eyebolt, h) Turnbuckles, i) U-clamp, j) Anchor fasteners (10 mm), k) Vertical PVC net cover, l) Periodical inspection & maintenance for tensioning, corrosion, damages and replacement etc. Selection of barricading system (Pipe railings or Wire Rope or combination suitable for the building profile) will be at the discretion of the Employer Project In-Charge.
ii	Lift shafts and other services shafts Protection	**Face side of shafts shall be closed by MS Frame**: **MOC: MS frame made of 35mm MS angle and plate, GI wire mesh-25x25x3mm properly welded & capable to hold lateral load of 100kgf.** a) Height of MS gate 1.80m minimum b) Width of MS grill: Based on the shaft opening size, with 6” more at both side for fastening. c) All joints welded and checked. d) Fixing by anchor fasteners. e) Entire opening shall be fully closed. f) Neatly painted in red & white bands. g) Warning signages shall be displayed on each lift shaft gates in visible font. h) Periodical inspection & maintenance for corrosion, welding defects, damages and replacement, repainting etc. **Horizontal closing of Shafts at Intervals to Arrest Free Fall**: **MOC: Suitable strength of safety net embedded with MS frame made of 40NB pipes or 50mm MS angle, flushed with shaft walls without gaps.** a) capable to hold vertical load of 1000kgf. b) Slope with 30 degrees towards landing. c) Fixed at interval of every 2 floors to arrest free fall of person/ material through the shaft.

S. N.	Fall Protection Requirements	Specifications & MOC for Safety Barricades and Railings
		Lift shaft working platform during the RCC work: All lift shafts shall be fully decked at working/live floor and de-shuttering floor by installing Self-Climbing platform in all lift shafts, covering two levels at least. **MEP & Other Services Cut-Outs**: All openings shall be protected horizontally and vertically by providing: a) Horizontal MS Mesh (35mm angle frame and inside mesh of 12mm rebar, spacing not more than 2" properly welded) and Standard hard barricading. b) All external chajja's of kitchen shafts, toilet shafts etc shall be protected by MS mesh fixed on openings. c) Edges of kitchen shafts & toilet shafts to be secured against fall by installing the standard hard barricading with toe board. d) **Larger Floor Cut Outs/Openings** (More than 3 feet) shall be secured with two layer of protections- Standard hard barricading with toe board and horizontal covering of opening with MS Mesh (35mm angle frame and inside mesh of 12mm rebar, spacing not more than 2" properly welded and supported by bracings) e) Periodical inspection & maintenance for corrosion, welding defects, damages and replacement, repainting etc.
iii	Excavation and Pilling Work Fall Protection Railing	**MOC: Minimum 40NB MS pipes, scaffold clamps & baseplates**: a) Top horizontal guardrail at 1150mm height b) Intermediate guardrail at 550mm c) Vertical posts at not more than 3m apart. d) For vertical posts: Base plate 150 x 150 mm, rigidly fixed on floor by anchor fasteners or grouted or 600 mm grouting in earth. a) Connector for railing pipes: scaffold clamps/ nut-bolts/ welding. b) Toe-board of 150mm height. c) Strength of railings: Capable to bear lateral load of about 100kg. d) Neatly painted in red & white bands e) For open excavations railings shall be fixed 1m away from the edge of excavation.

S. N.	Fall Protection Requirements	Specifications & MOC for Safety Barricades and Railings
		f) Other shallow excavations shall also be barricaded by similar railing protection to prevent fall of person or vehicle. g) Bored piling holes shall be protected against personnel falls by covering them with mild steel (MS) mesh, fabricated from 12 mm diameter rebar, properly welded with a spacing not exceeding 2 inches until concrete is poured. h) Periodical inspection & maintenance for corrosion, welding defects, damages and replacement, repainting etc.
iv.	Signages and visibility Colour	a) For high visibility, the railings and barricades shall be coated with anticorrosion and then by red & white paint. b) Display "Danger" sign board of 150 x 150mm
v.	Mandatory: Contractor shall submit fall protection plan, tracker in excel sheet, scheme of barricades, Scaffoldings, & safety screen and fabrication schedule for approval of the Employer Project In-Charge.	
vi.	Leading edges of constructed floors, top working floor leading edges, staircases; voids, MEP cut-outs, services shaft cut-outs, lift shafts, open balconies, temporary platforms, scaffold railings, suspended platforms, elevated platforms, projected decks etc, connecting walk-way beams, refuge floors slab edges etc.	MOC- Minimum 40NB MS pipes, scaffold clamps & baseplates, fasteners and vertical monofilament/ PVC net cover: a) Top horizontal guardrail at 1150mm height b) Intermediate guardrail at 550mm c) Vertical posts at not more than 3m apart. d) Vertical Posts: Base plate 150 x 150 mm, rigidly fixed on floor by anchor fasteners (10 mm). e) Connector for railing pipes: scaffold clamps/ nut-bolts/ welding. f) Toe-board of 150mm height. g) Strength of railings: Capable to bear lateral load of about 100kg. h) Neatly painted in red & white bands. i) Periodical inspection & maintenance for corrosion, welding defects, damages and replacement, repainting etc.
vii	Lift shafts and other services shafts Protection	MOC- MS frame made of 35mm MS angle or 16mm steel reinforcement, inside mesh of 10mm rebar spacing not more than 2" or GI wire mesh-3mm dia. 1"x1" spacing single braded fencing MS mesh, properly welded & capable to hold lateral load of 100kgf. a) Height of MS grill 1.80m minimum b) Width of MS grill: Based on the shaft opening size, with 6" more at both side for fastening. c) All joints welded and checked. d) Fixing by anchor fasteners.

S. N.	Fall Protection Requirements	Specifications & MOC for Safety Barricades and Railings
		e) Entire opening shall be fully closed. f) Removal of MS grills by Permit authorised by Project Manager of Contractor and Safety Manager of The Employer. g) Contractor may use waste steel.

16.2. **Safety Windscreen and Safety Net Protection**

The Civil contractor during RCC & related work shall provide adequate protection with combination of Safety Wind Screen and horizontal safety catch nets as per building height & design to control the risk of fall of material or person as per following mandatory structured requirements. The safety nets being used shall be of fire-retardant quality as approved by the Employer.

The Contractor shall submit fall protection plan of safety wind screen and netting for approval of the Employer Project In-Charge.

S. N.	Requirements	Specifications
i	Safety Windscreen: Fall protection for Tower covering. (i) Shuttering floors, de-shuttering floors and re-prop floors (top three operating floors) (ii) Void openings of tower and (iii) link beams of the tower.	Perimeter vertical safety wind-screen installation, made of MS frame of adequate strength and perforated MS screen sheets, climbing type hydraulically operated, covering all three top floors, flushed with peripheral working platforms. Similar protection with same windscreen shall be installed for voids and link-beams. The windscreen should support entire project cycle and should be installed at early stage of the project. The scheme for installation, jumping, periodic maintenance & inspection shall be approved by the Employer Project In-Charge and Safety Head. The contractor shall ensure annual fitness inspection, repair, replacement and stability certification from OEM.
ii	Protection below operating floor/ safety screen:	Safety catch nets- fan deck type 10m x 5m or suitable size just below operating floor/ safety screen but not more than 6m.
iii	Catch net protection at intermediate levels along the building height	Safety catch nets of suitable design and size of adequate strength at intervals of every 10 floors as per fall hazard risk and site requirement.
iv	Peripheral Protection for ground level	All along the periphery of Tower extending 10m from face of building and erected at height not more than 5m above base of the building or at entrance slab level.
v	Prevention of fall of objects below level- Net protection	All leading-edge safety railings towards the external face of tower shall be vertically flushed by

S. N.	Requirements	Specifications
		monofilament fire-retardant nettings/ PVC Net (Mesh Size 25 mm)
vi	Voids/ OTS/ Shafts	Horizontal safety catch nets fixed in the MS frames spread across length & breadth flushed with wall to wall at alternate five floors.
vii	Protection for working in external face of tower at height, external shafts, plumbing shafts, projected scaffolds/ platforms, projected/ open balconies, slab edge of floors, landings of open stairs etc and other high-risk work at height.	Vertical safety netting should be effectively used to envelope vertically the vulnerable work fronts to contain fall of material or person.
viii	Vertical Enclosure for High Rise Towers (applicable where Project located in proximity of the neighbouring occupied buildings, public roads, internal roads commercial property, public gardens etc)	In addition to above requirements, the Contractor shall provide additional protection safety measure to prevent fall/spillage of construction material, debris, tools etc on the neighbouring properties during project construction cycle. Any such possibility of impact on third party shall be prevented by providing appropriate vertical safety cover all along height of under construction tower, and/or specially designed horizontal safety net protection shielding the neighbouring community/property. It should serve the purpose of arresting projectiles of likely falling material/debris/tools etc during various construction activities and post pour activities. The contractor should submit appropriate scheme for approval of the Employer Project In-Charge and install it from beginning of the RCC work with ease of operating RSP during finishing phase. The Contractor shall be responsible to install it, inspect periodically and repair in case of damages through competent agency. The vertically enveloping material/safety net shall be fire retardant, with robust support structure and should withstand the weather condition and wind pressure.

16.3. Horizontal catch nets shall be erected at an angle of not more than 20^0 to its horizontal sloping into the building and secured with building structure or appropriate anchoring mechanism, but shall not be tied with any temporary structure like scaffold, CT Prop, shuttering, temporary railing etc.

16.4. In addition to hard barrication, wherever required or in case where hard protection is not practically possible, vertical safety nets shall be effectively used to envelope the vulnerable work fronts to contain fall of material or person from the building. (Rule 68 & 206 MBOCW).

16.5. The Contractor shall submit the safety net scheme to The Employer Project Manager and The Employer Safety Manager for approval. Specification of safety nets shall be as per IS-11057:1980 or as prescribed by the Employer.

16.6. Safety nets and its support structure shall be drop-tested after completion of installation, and thereafter every six months interval and or after major repair or relocation.

16.7. Whenever the existing barricades or railings need to be removed, permit shall be obtained with alternate precautions & those removed barricades shall be placed back while the activity is completed before closure of permit.

16.8. The contractor shall ensure fall restrain system tied in separate lifeline before deploying workmen at floor edge & at service shaft for internal painting & wall putty activity

16.9. Fire retardant vertical netting shall be provided to the external face of the building at working location to prevent fall of material & to limit emission of dust to the environment.

17. Lifelines & Fall-grab Arresters

17.1. While working with rope suspended platforms (RSP), mast-climber platforms, cantilever platforms, scaffolds, tower crane access ladder, work in lift-shafts & other shafts, external face of building etc the Contractor shall mandatorily ensure that the independent lifeline of 12mm/ 14mm Polyamide rope of approved quality and fall-grab arrester is provided. Ensure that the lifelines are independently fixed to the building structure or any other rigid support and not tied with the equipment/ platform on which the person is working.

17.2. Apart from above, contractor should maintain one set of emergency rescue kit, especially for the rescue of tower crane operator and other similar event/situation where person is suspended at height on safety harness due to sudden fall.

18. Working Platforms and Ladders

The Contractor shall ensure that proper working platform are provided for finishing work on floor level and MCA area. The platform shall be light weight modular aluminium mobile scaffolds with adjustable platform with top & mid rail and toe-boards, heavy duty caster wheels with lock, access ladder etc with minimum loading capacity of 200 kg/m3 with required out-riggers for stability (to prevent toppling). The platform scaffolds shall be periodically inspected for fitness and tagged. The A-Type ladders being unstable and unsafe shall not be allowed at site, instead ladder with platform and safety railing shall be used.

19. Scaffolding Safety

The Contractor shall provide scaffolding of following minimum specifications:

19.1. Heavy-duty scaffolding shall be used.

19.2. Scaffolding of four meters above height shall have tie support with permanent structure at interval of 4 meters.

19.3. Contractor shall deploy competent scaffolders for erecting and dismantling of scaffold.

19.4. Contractor shall deploy certified scaffolding foremen, who will be responsible to inspect and provide fit to use-tag (Scaff-tag).

19.5. Ladder shall be used for scaffold access.

19.6. Working platform of scaffold shall be fully covered by using transom and metal plank, or hook type metal boards. Challi platform is not recommended, which is vulnerable to stability. Also, platform should not be cantilever type extended beyond 150mm.

19.7. Top handrails, mid rails and toe boards shall be fixed on scaffolding platform as per standard.

19.8. Scaffold shall be erected over firm and plain surface. In case ground surface is not levelled then sole plate shall be used after levelling.

19.9. Scaffold shall be erected over base plate over firm levelled surface or over sole plate having width thrice than base plate.

19.10. Scaffolding erected having more than 4m height shall be as per approved scheme.

19.11. Defective scaffold like bend, cracked, rusted etc shall not be used.

19.12. Cross bracing shall be ensured for all scaffolds and for height above four jumps, diagonal bracings shall be given.

20. Incident Reporting and Investigation

The contractor shall report all type of incidents to the Employer Project In-Charge as per SOP-1 Incident Reporting, Investigation & Recording Procedure, and timeline shall be adhered as mentioned in the SOP, which is also listed below. The FIR report should be followed by investigation report and corrective/preventive plan. Hiding or non-reporting of incident shall treat seriously and will attract heavy penalty and suspension of work. In the event of unfortunate incident & injury, it should be contractor's prime responsibility to provide immediate medical assistance to the victim of incident, any such delay or avoidance of responsibility by the contractor shall be dealt strictly.

Type of incident	**Timeline for Report Submission**	
	Incident Reporting	**Investigation Report Submission**
First Intimation Report (FIR) for all type of incidents	Immediately	Within 72 hours except Fatal
Fire incident and dangerous occurrence	Within 4 hours	Within 72 hours
Loss time incident (LTI)	Immediate, and detail report within 72 hours	Within 72 hours
Fatality/ disability	Immediate, within 2 hours	Within 48 hours

21. Safety Trainings

21.1. **Safety Induction of Employees:** It is mandatory that the contractor shall impart effective induction to all employees including staff and workers before reporting at workplace. The contractor shall have dedicated safety induction room equipped with audio-visual, safety film for screening, display safety policy, safety rules and instructions in language understood by workers. The safety officer of contractor shall be responsible to impart training, which should include explaining general safety rules of The Employer, probable construction hazards, risk, emergency & medical facility, fire protection & prevention, fall protection measures, electrical safety, safe rigging practices, use of PPEs, fire extinguishers, safety harness, safety nets, assembly point etc. The duration of safety induction shall be minimum two hours and record should be maintained.

21.2. **Work Related Trainings:** It is mandatory that the contractor shall plan monthly safety training calendar according to the activities planned, hazard and risk identified and safety control measures applicable to all employees including staff and workers before engaging at workplace.

21.3. **Toolbox Talk Training:** Due to literacy issue and social background of labours, daily sensitisation of the workers for safety & health adherence is mandatory. The Contractor shall conduct toolbox talk training of workers on daily basis in groups or tower-wise or activity wise. A mass TBT once in a month is must. The TBT should cover work related topics like hazard & risk, use of PPEs and tools, housekeeping, hot work, electrical safety, use of safety harness for height work, health & hygiene etc. TBT shall solicit active participation of workers and site engineers, and record shall be furnished to the Employer on weekly basis.

21.4. **Safety Awareness Campaign:** The Contractor shall at his cost conduct monthly safety motivational and awareness campaigns. Also organize the celebration of important Safety events (National Safety Week,

World Environment Day, Fire Service Week, Road Safety Week etc.) for staff and workers and motivate them suitably.

22. Safety Signages

22.1. The contractor shall display adequate numbers of signages, hoardings, posters, policy, rules, instructions etc. for safety awareness, including cautionary & danger and other informative signages as per site hazard and risk. They should be legible, prominently and placed at conspicuous places in the language understood by majority of the workers.

22.2. The contractor shall maintain these signboards and posters in good condition throughout the contract period and replace them periodically.

22.3. Contractor shall adequately display the posters, instruction boards and signages such as mandatory PPE pictorials, No Smoking signs, Flammable danger sign near fuel storages, DG sets, Caution for moving parts of equipment viz. fly wheels of piling rigs, motors of pumps, Scaffolding Scaff-tags, Danger sign- deep excavation, floor openings, electrical danger sign, vehicle speed limit, fire extinguishers, exit signs, assembly sign, debris falling caution, height work & safety harness etc., whichever applicable to them.

22.4. Develop Safety Park with display of policy, rules, best safety practices, all types of PPE and donning instructions.

23. Welfare of Building Workers

23.1. **Drinking Water**

i. The contractor shall provide adequate drinking water facility, which includes sufficient water taps, water purifiers and water coolers their staff and workers at his own cost.

ii. Drinking water tanks should be cleaned fortnightly or as required and potability test should be done six monthly (as per IS:10500) and should be equipped with appropriate filtration media.

iii. Also ensure that water tank lids are closed or follow directives of local Municipal Corporation, to prevent the larva growth and mosquito breeding etc.

iv. Signage with pictorial shall be placed for drinking water in English, Hindi & local language.

23.2. **Crèche:** The Contractor shall provide crèche at site, if more than 50 female building workers are employed and it shall be under the charge of women trained in the care of children & infants and maintain in sanitary condition and adhere with Section 35 of BOCW Act.

23.3. **Canteen**

The contractor shall provide and maintain wherein more than 250 workers are employed, a canteen for the use of the workers and adhere with section 37 of BOCW Act.

i. Canteen shall be situated at the distance not less than 15.2 meter away from any latrine/urinal or any source of dust, smoke or fumes and should be equipped with fly catcher. Also waste water from canteen shall be carried away in covered drain.

ii. Arrangements shall be made for the collection & disposal of canteen & labour camp food waste daily.

iii. Medical examination of canteen food handlers shall be done twice in a year (if applicable).

iv. At site provide & identify lunch shed with drinking water facility& dust bin. Eating food in workplace is not allowed.

23.4. **Accommodation**

The labour camp accommodation shall comply with the Employer Guidelines-4 - Temporary Labour Camp of Contractors Health, Hygiene, Welfare & Safety Requirements.

23.5. **Shelter**

The Contractor shall arrange suitable weather shelters for lunch and resting of the workers, at a safe distance away from construction activities.

23.6. **Labour Toilet, Urinals and Pest Control**

The Housekeeping and Hygiene at The Employer Projects is of utmost importance. The contractor shall ensure sanitation and hygiene at workplace and at Labour Camp. He shall submit the sanitation logistics plan for site and camp for approval of Project In-Charge of The Employer.

The latrines and urinals, as the case, shall be provided as specified below:

	For Male Workers	**For Female Workers**
Urinals		
1-50 workers	1 Urinal	1 Water-Closet per 50 female workers with separate entry & pictorial sign.
50-500 Workers	1 Urinal per 50 workers	
500 & above	Addition to above 1 additional urinal per 100 workers	
Inside the Towers by Civil Contractor	1 Urinal in the Tower after every two floors	-
Latrines		
1-25 Workers	1 latrine	1 latrine per 25 female workers with separate entry and pictorial sign & notice
25-100 workers	Additional to above 1 latrine per 25 workers	
100 & above	Additional to above 1 latrine per 100 workers	

i. Every latrine shall be under cover partitioned off to secure privacy with proper door and fastenings, adequately lighted, and maintained in a clean and sanitation condition at all times.

ii. Where both male and female building workers are employed, display outside each block of latrines or urinals a sign as “For Men Only” or “For Women Only” with pictorial sign of man or of a woman, as the case may be, written in the language understood by the majority.

iii. The contractor shall employ full time sufficient sweepers to maintain latrine and urinal blocks in clean and hygiene condition and maintain logbook.

iv. It shall be cleaned and disinfected at least every day and maintained properly by the contractor throughout the project duration.

v. Proper disposal of excreta by septic tank and soak pit shall be made by the contractor. In no case, the excreta shall be disposed off in any open drain, nallah, etc. which may cause outbreak of disease or affect overall hygiene of the workplace.

vi. The floor area should be of impervious material easy for cleaning & sweeping.

vii. Provision of sufficient illumination should be ensured.

viii. Pest control such as - larva treatment, mosquito fogging, rodent treatment, bed bugs, snake repellent shall be done weekly or as instructed by project safety manager.

ix. Whitewash of every latrines/urinal once in every period of four months.

24. Health Management and Medical Facility

The major contractor shall arrange adequate facilities for medical aid and treatment for their staff and workers including the first-aid facilities at the project site. In general, the contractor shall adhere to the Rule 250, 253, 254, 255, 256, 258, 259 and Schedule III of MBOCW Rules, 2007.

24.1. **First Aid Centre**

The contractor shall ensure first aid centre (ambulance room) at site as per Rule 253 MBOCW Rules to facilitate immediate relief to the injured person. The centre shall be under charge of qualified Nurse, who shall maintain first aid injury & treatment record. The centre should be equipped with the articles specified in Schedule IX of MBOCW Rules. The size of the room shall be adequate for proper treatment of the injured persons and keeping the enlisted articles in an organized manner. The room shall be well ventilated and illuminated, preferably by natural means. The contractor shall keep a refrigerator of approx. 150 litres capacity for proper storage of injections and temperature sensitive medicines etc.

24.2. Preliminary **Medical Examination of Workers**

The Contractors shall at his cost conduct preliminary medical examination of workers before deploying at site and thereafter once in a year and submit fitness certificate to The Employer and ensure that the person having defective vision, deafness & has a tendency to giddiness are prohibited from employment. For height work, persons should be subjected to vertigo test and records should be available.

24.3. **Medical** Examination **of Operators and Drivers**

The Contractor shall arrange mandatory medical examination of persons employed as drivers and operators such as lifting appliances, plant & machinery, transport vehicles, crane operators & signaller, mobile cranes, placer-booms, rope suspended platforms (RSP-gondola) & passenger-cum material hoist, pilling machines, electricians, technicians, batching plant operating staff, height work access platforms, before deploying at work, which shall include full medical and occupational history, clinical examination which includes general physique, vision, hearing, breathing test, upper & lower limb function, spine, general mental alertness and any other necessary test recommended by examining doctor. The medical fitness certificate shall be submitted to the Employer in Form XXVIII.

24.4. **Medical** Examination **Record**

The contractor shall keep record of medical examination of every building worker employed by him in a health register in Form XXVIII MBCOW till end of the project.

24.5. **Hospital Tie-Up for Medical Emergency**

The Contractors individually or collectively shall have official tie-up with the nearest well-equipped hospital for handling medical or accidental emergency at site.

24.6. Ambulance **Van**

Ambulance should be available any time at site or on call for emergency transportation. Telephone number & address of such hospital, doctor, ambulance services shall be prominently displayed. A copy of agreement with hospital shall be furnished to The Employer.

24.7. Medical **Officer**

The major contractor shall engage a medical officer as per Rule 256 MBOCW to provide medical consultation & treatment to the workers. The doctor's name, contact number and visiting hours shall be displayed prominently.

24.8. Stretcher

The Contractor shall keep minimum two stretchers in ready to use condition.

24.9. **First** Aid **Box**

The Contractor employing more than 10 workers shall have first-aid box available under the charge of trained first aider with signage as "First Aid Box Available". The contents of first aid box should be as per Schedule XII of MBCOW.

25. Passenger Hoists and Material Hoists

25.1. The contractor installing the passenger-cum-material hoists for their use or the vendor providing such hoists for The Employer use on hire basis shall ensure comprehensive third-party tests & examination of the hoists by the Competent Person before being taken into use after installation or every shifting and there after thorough examination every six month and submit test certificate to the Employer Project In-Charge. Unless this is done the hoists shall not be used.

25.2. The Hoists case shall be so constructed as to prevent, when its gates are shut, any building worker carried by such hoist from falling out of it or from being trapped between any part of such cage and any fixed structure or other moving part of such hoist or from being struck by articles or materials falling down the hoist-way on which such hoist is moving.

25.3. The Hoists case shall be fitted on each of its side from which, access is provided to a landing place with a gate which has efficient interlocking or other devices to secure so that such gate cannot be opened except when such cage is at a landing place and that such cage cannot be moved away from any such place until such gate is closed.

25.4. Every gate at landings and hoist-way enclosure of such hoist is fitted with efficient interlocking or lock & key or other devices to secure so that gate cannot be opened except when the cage of hoists is at the landing place, and that such large cage cannot be moved away from the landing place until such gate is closed.

25.5. In every hoist used for carrying building workers there are provided suitable and efficient automatic devices to ensure that the cage of such hoist comes to rest at a point above the lowest point to which such cage may travel.

25.6. The passenger hoist shall be equipped with a wireless floor calling system.

25.7. The approach to a hoist is adequately lit, fire extinguisher provided in hoist cage, safety operations instructions are displayed, and emergency ladder is available.

25.8. Display SWL prominently in the cage. Display maximum number of persons to be carried and combination of material and person to be carried.

25.9. No person other than the operator trained, capable and medically fit to operate shall be employed to operate hoists.

25.10. The Contractor or vendor shall ensure that the preventive maintenance of the hoists is carried out as per original equipment manufacturers (OEM) manual & frequency and daily inspection is conducted by operator before using the hoists and record available.

25.11. Floor landing numbers along the hoist way shall be distinctively marked and legible.

25.12. Requirement of Rule No 92, 97, 100, 101 & 105 MBOCW Rules 2007 shall be complied.

26. Work Permit

All work having high risk of safety should be undertaken under proper “permit to work”. The following work permit should be compulsory to be used at The Employer construction site.

26.1. Hot work permit: Gas cutting, welding, brazing, heating, etc.

26.2. Height work permit: All types of work above 2m height from the floor level.

26.3. Confined space work permit: Working inside silo, tank, manholes & pipelines/pits or any other location where there is a risk of insufficient air circulation or restricted entry space, etc.

26.4. Safety facility removal permit: Removal of any type of safety protection facilities including barricades, MEP covers, safety nets, lifeline/wire rope barrication, periphery safety nets, safety screens, related signages, etc.

26.5. Electric work: Working on electric supply system, panels, heavy machines operated on electric supply, batching plant etc.

26.6. Night work: Any work carried out in the evening beyond normal working hours of the site
26.7. Excavation work: Any excavation work, trenching, cutting of earth beyond 2m deep
26.8. Demolition work: Any demolition work of standing structure or building

The permit shall be authorized by the Employer Site Engineer/location in charge as per the standard operating procedure. The Employer safety staff are authorized to verify the permit compliance and in case of violation/deviation, should stop the job till desired rectification is achieved.

27. Legal Compliance and Register

The contractor shall identify the requirements of good practices at site for fulfilment of legal requirements related to environment, occupational health and safety and welfare. The contractor shall enlist all the activities under the contract in advance in the Legal Register and their effect on safety, health and environment. The contactor shall establish the procedure to take all legal approvals and timely renewal in advance. The contractor shall maintain an update Legal Register and The Employer Safety Manager shall verify quarterly.

List of Legal act and rules.

27.1. Building & Other Construction Workers (Regulation of Employment & Conditions of Service) Act, 1996 and Maharashtra BOCW Rules 2007.
27.2. The Gas Cylinders Rules 2016
27.3. The Indian Electricity Act 2003
27.4. The Indian Electricity Rules 1956 & 2005
27.5. Fatal Accidents Act, 1855
27.6. Explosives Act, 1884 and Explosives Substances Act, 1908 & Explosives Rules, 2008
27.7. Motor Vehicles Act, 1988 and Central Motor Vehicles Rules, 1989
27.8. Petroleum Act, 1934 and Rules, 2002
27.9. The Static & Mobile Pressure Vessels [Unfired] Rules 2016
27.10. Noise Pollution Regulation & Control Rules, 2000
27.11. Environment (Protection) Act, 1986 and Rules, 1986.
27.12. The Hazardous Waste (Management & Handling) Rules, 1989
27.13. Air (Prevention & Control of Pollution) Act, 1981 and Rules, 1982
27.14. National Ambient Air Quality Standards- Notification 1994 under Air Act.
27.15. Water (Prevention & Control of Pollution) Act, 1974 and Rules, 1975
27.16. Battery (Management and Handling) Rules, 2001
27.17. Bio-Medical Waste (Management & Handling) Rules1998
27.18. Child Labour (Prohibition & Regulation) Act, 1986 and Rules, 1988
27.19. Contract Labour (Regulation & Abolition) Act, 1970 and Rules, 1971
27.20. Inter-State Migrant Workmen (Regulation of Employment & Conditions or Service) Act, 1979 and Rules, 1980
27.21. Public Liability Insurance Act, 1991 and Rules
27.22. Workmen's Compensation Act, 1923 and State Rules.
27.23. Minimum Wages Act, 1948
27.24. Payment of Wages Act, 1936
27.25. Employees Provident Fund and Miscellaneous Provisions Act, 1952
27.26. Employees State Insurance Act, 1948
27.27. Employee's Deposit Linked Insurance Scheme, 1976
27.28. Bombay Labour Welfare Fund Act, 1953

27.29. Maharashtra Contract Labour (Regulation and Abolition) Rules, 1971
27.30. Payment of Bonus Act, 1965
27.31. Maternity Benefit Act, 1961
27.32. Payment of Gratuity Act, 1972
27.33. Compliances list as per Annexure 6 of the Works Contact: List of Statutory Compliances; and
27.34. Any statutory modifications or re-enactments thereto.
27.35. National Green Tribunal (NGT) Act 2010

The contractor shall display abstract of prominent safety and labour laws at conspicuous places at site.

28. Penalty Against Safety Violations and Accident / Incident

The contractor shall comply with all the requirements laid down in the contract agreement-safety rules and regulation and BOCW Act 1996 and MBOCW Rules 2007, and any other safety & health requirements as a matter of general judiciousness. The safety violations leading to any type of incident and injury are not acceptable to the Employer and this will attract heavy penalty immediately at the discretion of Project In-Charge of the Employer including termination of the contract or removal of defaulting employees. The Employer Project In-Charge shall communicate its decision to defaulting contractor officially and proper record shall be maintained, and penalty shall be deducted from contractor's next running bill. The nature of violations and the penalties are more particularly described below:

28.1. **For Category Level 1,2 & 3**

i.First Safety Violation: Warning
ii.Second Safety Violation: Penalty as per schedule
iii.Third Safety Violation: Double penalty
iv.Fourth Safety Violation: Dismissal from site

28.2. **For Level 4 Category**

i.First Incident: Penalty
ii.Second Incident: Dismissal

S. N.	Nature of Safety and Housekeeping Violations	Penalty in Rs. including applicable GST (per violation)
Level 1 Penalty		
i	Mandatory PPE's not used by employees of Contractor or not provided by Contractor (Safety helmet, safety shoes and reflective jacket).	Rs. 500/-
ii	Other job specific PPE (hand gloves, safety goggles, welder goggles, dust mask, ear plugs etc.) not provided or used by employees of Contractor.	Rs. 500/-
iii	Smoking in site premises (fire, explosion & health hazard)	Rs. 500/-
Level 2 Penalty		
iv	Contractor's worker or staff working without mandatory safety induction (site orientation on hazards & incident).	Rs. 2000/-

S. N.	Nature of Safety and Housekeeping Violations	Penalty in Rs. including applicable GST (per violation)
v	Haphazard/ obstructive material stacking at workplace, passages and in yards and fall of unsecured tools/ objects (trip, fall or struck-by hazard)	Rs. 2000/-
vi	Inadequate up-keep at site and labour camp, unhygienic toilets, bath area, drinking water, fire/explosion hazard etc (health hazard & fire hazard)	Rs. 2000/-
vii	Use of unsafe & unguarded power hand tools such as cutter, grinder etc. (Amputation risk)	Rs. 2000/-
viii	Use of unfit vehicles (reverse horn, mirror, cabin gate, flood lights not provided, daily inspection missing) (vehicular incident).	Rs. 2000/-
ix	Electric cables trailing on walkways, routed on floor, metallic structures, water (tripping hazard/ electrocution hazard).	Rs. 2000/-
x	Unsafe electric connections- damaged cables, multiple-joints, bare wire connection without 3-pin-plugs, multiple connections from single socket etc. (electrocutions, short circuit/ fire hazard)	Rs. 2000/-
xi	Improper storage and unsecured handling of gas cylinders without wheel mounted trolley (toppling/ roll-down of cylinders, fire & explosion hazard)	Rs. 2000/-
Level 3 Penalty		
xii	First aid kit not available with contractor (delayed medical assistance)	Rs. 10,000/-
xiii	Working at height without safety harness or safety harness not anchored (fall hazard/ fatality risk)	Rs. 10,000/-
xiv	Working at external face of towers and shafts without lifeline & fall grab arrester (fall hazard/ fatality risk)	Rs. 10,000/-
xv	Poor housekeeping, debris, combustibles, excess construction accessories etc. not removed from workplace (trip/ fall hazard or fire hazard)	Rs. 10,000/-
xvi	Debris/ construction material openly thrown from height, towers (bodily injury to several people)	Rs. 10,000/-
xvii	Hot work without fire extinguisher, fire blanket and permit or de-pressurised fire extinguisher (fire hazard/delay in fire mitigation)	Rs. 10,000/-
xviii	Safety catch net, fall protection barricading of trenches, excavation, shafts, MEP openings, slab edge not provided or removed without permission of the Employer Project In-Charge/Safety Manager (fall of material/ person hazard).	Rs. 10,000/-
xix	Unsafe working platform, scaffold, shuttering etc. (fall of material/ person hazard)	Rs. 10,000/-

S. N.	Nature of Safety and Housekeeping Violations	Penalty in Rs. including applicable GST (per violation)
xx	Electric DB without 30mA RCCB/RCCB, non-standard design and earthing connection (electrocution hazard)	Rs. 10,000/-
xxi	Passenger hoist operation without competent operator, unsecured landing gates & locking etc (fall hazard/crushing risk)	Rs. 10,000/-
xxii	Using material lifting tools, tackles, hoists, gondola & cranes without TPI (risk of machine failure/ fall of material/ incident)	Rs. 10,000/-
xxiii	Operating earth moving vehicles without licensed & competent driver (un-authorised driving/ risk of vehicular incident)	Rs. 10,000/-
xxiv	Violation to BOCW Requirements (legal threat to the Employer, risk of incident)	Rs. 10,000/-
xxv	Deviation to work method statement and risk assessment control measures (risk of incident & quality issue)	Rs. 10,000/-
xxvi	Safety staff, safety officer, safety supervisor not deployed as per contract condition (lack of safety management and supervision)	Rs. 10,000/-
xxvii	Not complying with safety observations & instructions, third party audit observations etc. on time and evidence of repetitive violations.	Rs. 10,000/-
xxviii	Not attending project safety committee meeting (delay in decision & communication)	Rs. 10,000/-
xxix	Not reporting near-miss and incidents within 24 hrs to the Employer Project In-Charge or safety manager (Deviation to regulation & delay in corrective/ preventive action).	Rs. 10,000/-
xxx	Any other safety violation not listed above; the Employer Project In-Charge shall levy penalty not amounting to more than Rs.10,000/-	Rs. 10,000/-
Level 4 Penalty		
xxxi	Fire incident	Rs. 50,000/-
xxxii	Leakage / spillage of flammable or toxic gas & liquids	Rs. 50,000/-
xxxiii	Medical treatment case (MT) due to negligence	Rs. 50,000/-
xxxiv	Dangerous Occurrence with high potential as defined in MBOCW Rules	Rs. 50,000/-
xxxv	Lost time reportable incident (LTI)	Rs. 1,00,000/-
xxxvi	Fatality/ disability	Heavy Penalty and or disciplinary action as deemed fit basis investigation.

29. Additional Safety Rules & Regulations

The Employer reserves the right to issue additional safety rules and regulations, standard operating procedures, work instructions, guidelines and checklists as and when required particularly under exceptional hazardous working conditions, new plant & machineries, change of operational conditions/ methods, construction of high-rise towers, location & neighbouring constraints of construction site etc and same shall be binding on the contractor to adhere and ensure additional safety precautions for safety of people and property at his time and cost.

30. Abbreviations

BOCW Act: The Building and Other Construction Workers (R E & C S) Act 1996.
DB: Distribution board
LPG: Liquified petroleum gas
LOTO: Lock-out and Tag-Out
LTI: Lost time reportable incident
MBOCW Rules: Maharashtra Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Rules 2007.
MT: Medical treatment injury
MEP: Mechanical, electrical and plumbing
PPE: Personal protective equipment
PTW: Permit to work
RCCB: Residual current circuit breaker
SMART: Specific, measurable, achievable, realistic and time-bound
TBT: Toolbox talk
ELCB: Earth leakage circuit breaker

31. Confined Space Safety

31.1. The contractor shall ensure extreme care while working inside any type of confined space from inhaling dust, fumes, gases, vapours (e.g. water tank, STP tank, vessels, Silos, pit with limited access, drainage man-holes, water-proofing work inside tanks, painting/special chemical coating work inside tanks, containers with limited man-hole access, shafts with limited access, cleaning/ washing of tanks, maintenance work inside tanks or erecting of scaffolds/ladders inside tanks, chimney, oil tanks, grinding, cleaning, spraying etc). All such confined space should be accessed only after obtaining all clear "Confined Space Entry Work Permit". Before entering any such confined space ensure all manholes are adequately ventilated and Oxygen deficiency has been checked by instrument and proper access ladder and watch-guard available.

31.2. Special precautions are must for any entry into Sewer tanks, lines, septic tanks, maintenance, water proofing, painting inside tanks etc, which are already in use. In such cases ensure all manholes are open, adequately ventilated for sufficient time, before entry atmosphere inside is checked for the presence of any explosive mixture, toxic gas or oxygen deficiency. The workers shall be allowed to get into the space after obtaining Permit and supervision of watch-guard only.

31.3. The manholes opened shall be cordoned off by suitable railing, caution signage, proper illumination to prevent accidental fall.

31.4. Forced ventilation system shall be utilised for working inside all type of Confined space with battery backup and emergency rescue measures for workers shall be ensured.

31.5. During water proofing work inside the tanks appropriate time gap to be maintained to ensure limited exposure.

31.6. All safety measures for working in confined space as given in BOCW Act-Section 40 and MBOCW- Rule 67 shall be complied.

32. Diesel Storage and Flammable Material

The Contractor intends to store Diesel, Paint, Thinner or any other fuel/ flammable liquids shall adhere to the requirements of The Petroleum Act, 1934 and Rules, 2002 and IS 7969 (1975): Safety code for handling and storage of building materials including following specifications:

32.1. Covered Storage area with PCC and sufficient slope towards collection pit. Provide bund of 110% of stored volume of fuel. Ensure adequate ventilation to prevent accumulation of flammable vapours.

32.2. As far as possible, fuel drum shall be placed over drip tray.

32.3. No light fixture other than flame proof light to be fixed inside the storage area.

32.4. No electric connection shall be installed inside the storage area.

32.5. Hand pump shall be used for fuelling vehicles or transfer from one container to another.

32.6. Foam type and DCP type extinguisher of adequate capacity shall be placed outside the storage area with location signage.

32.7. No Smoking, No Hot Work, no open Flame signage shall be prominently displayed outside.

32.8. MSDS and immediate first aid measures shall be displayed.

32.9. For storage of Class-B petroleum i.e. Diesel exceeding 2500liters, license shall be obtained.

32.10. Paints, varnishes, lacquers, thinners and other flammable materials shall be kept in a properly sealed or closed containers.

32.11. The container shall be kept in a well-ventilated location, free from excessive heat, smoke, sparks or flame. Constructed from non-combustible or fire-rated materials.

32.12. Flooring must be impervious (e.g., epoxy-coated concrete) and with spill containment bunds.

32.13. Store incompatible chemicals separately (e.g., acids vs. bases, oxidizers vs. solvents).

32.14. Use dedicated cabinets for flammable, corrosive, or toxic chemicals, grounded to prevent static build-up.

32.15. Never store paints with food, water, or drinking supplies.

32.16. No Smoking, No Hot Work, no open Flame signage shall be prominently displayed outside.

32.17. MSDS and immediate first aid measures shall be displayed.

32.18. Buckets containing sand shall be kept ready for use in case of fire. Fire extinguishers, when required, shall be of foam type.

33. Ambient Air & Noise Pollution Monitoring

The Contractor shall conduct ambient air & noise monitoring survey through authorized agency at initial stage of the project and later every six months as per State Pollution Control Board / CPCB Standards and ensure necessary arrangement for control/ elimination of pollution and suppression of dust by spraying water on dusty roads and affected areas. The noise pollution shall be controlled by effective engineering control. The Contractor shall adhere with Rule 61, 67, 70, 79 of MBOCW Rules. Shall submit copy of report to The Employer.

34. Local Governing body statutory requirement for Air Pollution Mitigation

34.1. All the project proponents to ensure that at least 35 feet high tin / metal sheets shall be erected around the periphery of construction projects having height more than 70 mtr.

34.2. All construction layouts having area more than l(one) acre shall have tin / metal sheet erected of height 35 feet at least around periphery of the construction project sites and for construction sites, less than 1 (one) acre, the tin / metal sheet height shall be 25 feet at least.

34.3. All the buildings under construction shall be compulsorily enclosed by monofilament fire retardant safety net from all sides.

34.4. It shall be ensured that water fogging shall be carried out during loading and unloading of materials at the construction sites (use of stationary/ mobile anti-smog guns)

34.5. The water sprinkling shall be done on debris, earth material etc. which are prone to generate airborne particulate matters at all construction sites without fail.

34.6. All vehicles carrying construction materials shall be fully covered (i.e. from top and all sides) so that construction material or debris does not become airborne during transportation and the vehicle shall not be overloaded to avoid any spillage from the vehicle

34.7. All construction sites should install CCTV cameras along the periphery of their work sites to ensure that vehicles are playing after cleaning tyres and are not overloaded

34.8. All construction sites deploy sensor-based Air Quality Monitors at construction sites and act immediately on observing pollution levels exceeding the limits.

34.9. If such devices are not installed action will be taken against such units including to close down such units till compliance is achieved.

34.10. The installation of these sensor-based air quality monitors and their integration with the Central Command Centre will enable municipal corporations to track real-time pollution levels at the source and take necessary action.

34.11. The construction site shall also display real-time pollution measurement data visibly at the site on LED screen for public. This sensor-based air quality monitors should feature a satisfactory protocol to transmit the data to the Central Command Centre of municipal corporation for regular monitoring and supervisory control.

34.12. Sensors must possess certification such as mCERTS TUV, NIST, or AirParif, or have undergone performance evaluations by institutions like IIT Kanpur, IIT Bombay, IIT Delhi, IIT Madras, IIT Kharagpur, or CSTEP Bangalore.

34.13. All the work sites shall ensure that the grinding, cutting, drilling, sawing and trimming work is carried out in enclosed areas and water sprinkling / water fogging is continuously done while working to avoid escape of fugitive air.

34.14. All the construction sites shall ensure that C & D (Construction and Demolition) waste generated within the premises / site of work is transported to designated unloading site strictly as per municipal waste management plan. After unloading the debris, the vehicle shall be washed and cleaned thoroughly.

34.15. All vehicles carrying materials shall have valid PUC certificates and the same shall be produced as and when asked for by competent authorities.

34.16. The loose soil, sand, construction materials and debris of any kind and quantity shall be stored in demarcated / dedicated area and properly barricaded, fully covered / enclosed / protected with tarpaulins. It shall be ensured that there is no dumping of construction material and debris on public roads, footpaths pavements and open area.

34.17. Vehicle tyre washing facility shall be provided at all exit points of construction sites. It shall be ensured that daily cleaning is carried out of major roads for removal of dust by using vacuum sweeping or water sprinkling, brushing brooming and sweeping.

34.18. There shall be complete ban on open burning anywhere in the geographical area under municipal corporation.

35. Environment, Health & Safety Plan of Project (EHS Plan):

35.1. **Concrete, Shuttering, Rebar**

The Contractor has both legal and contractual responsibility to formulate and implement Scope specific EHS Plan before proceeding for any construction activity. EHS Plan shall be prepared as per The Employer guidelines and approval of the Employer Project In-Charge is mandatory. The emergency management plan as per MBOCW (Rule 63), shall be part of the EHS Plan. The Contractor should submit EHS Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

35.2. **Demolition of existing structure**

The Contractor has both legal (chapter XVI of MBOCW Rule 2007) and contractual responsibility to formulate site-specific Demolition Plan for approval before work starts. This may include but not limited to,

i. Structural analysis of the building/ structure to be demolished
ii. Sequence of demolition
iii. Equipment and methods to be used including control blasting, if applicable
iv. Safety measures for personnel and the public in the vicinity
v. Confirm isolation of all existing utilities: PNG gas, electricity, water, sewer, telecom and other applicable
vi. Asbestos and hazardous materials survey and abatement before starting
vii. Dust and Noise Control measures as per Environment (Protection) Act,1986 and Rules, 1986.
viii. Debris and other waste management process
ix. Complying with local Municipal Guidelines for air pollution mitigation.

Demolition plan shall be prepared as per The Employer guidelines and approval of The Employer Project In-Charge is mandatory. The emergency management plan as per MBOCW (Rule 63), shall be part of the Demolition Plan. The Contractor should submit Demolition Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

35.3. **Earthwork, Pile and rock anchor work, soil retention**

The Contractor has both legal and contractual responsibility to formulate and implement SMART Excavation Plan before proceeding for any excavation activity. This plan shall be prepared as per The Employer guidelines, specifications mentioned in IS 3764:1992 (Excavation work code of safety) and approval of The Employer Project In-Charge is mandatory. The emergency management plan as per MBOCW (Rule 63), shall be part of the EHS plan.

The excavation plan shall address key safety and operational aspects, including identification of underground services, impacts on adjacent structures, overhead hazards, unstable soil conditions, risks of cave-in, uneven terrain, access and egress routes, pedestrian safety, groundwater levels, surface runoff, vehicular traffic, lightning exposure, potential wildlife encounters and soil contamination, as applicable.

The Contractor should submit the Excavation Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

35.4. **Heavy** Fabrications **(Design and Build)**

The Contractor has both legal and contractual responsibility to formulate and implement SMART Lifting erection & structural fabrication plan before proceeding for any lifting & erection activity. This shall be prepared as per The employer guidelines and approval of The Employer Project In-Charge is mandatory.

This plan shall outline the procedures and safety measures for a specific lifting & erection operation, includes information about the shape of the structure, load, lifting equipment, personnel, and the lifting method, ensuring a safe and successful lift. Key elements of the plan shall include details about the load's weight, dimensions, and centre of gravity, as well as information about the lifting equipment, such as crane capacity, type, and rigging etc.,

This plan shall confirm the basic requirement mentioned in MBOCW rule 210 - Erection or dismantling of steel and prefabricated structure & IS:7205-1974 (Safety code for erection of structural steelwork)

The Contractor should submit the plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

35.5. **Flat Façade, Glass Railing, GRC/ GFRC, Other Façade**

The Contractor has both legal and contractual responsibility to formulate and implement SMART EHS Plan before proceeding for any construction activity. EHS Plan shall be prepared as per The Employer guidelines & shall include but not limited to - types of façade materials, lifting plan, transportation & storage arrangement, competency of riggers / operators, equipment's involved, maintenance & inspection schedule etc., and shall be approved by The Employer Project In-Charge. The emergency management plan as per MBOCW (Rule 63), shall be part of the EHS Plan. The Contractor should submit EHS Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

35.6. **Internal Electrical works, Substation**

The Contractor has both legal and contractual responsibility to formulate and implement SMART EHS Plan before proceeding for any construction activity. EHS Plan shall be prepared as per The Employer guidelines & shall include but not limited to – scope & type of activity like Electrical Conduiting & wiring, Lightning Arrestor, Aviation Light System, Substation & materials used for the activity, material storage arrangement, erection & installation, equipment's & machineries used etc. and shall be approved by The Employer Project In-Charge. The emergency management plan as per MBOCW (Rule 63), shall be part of the EHS Plan. The Contractor should submit EHS Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

35.7. Plumbing **water supply and drainage works, infra plumbing, firefighting, HVAC, IBS/ FTTH/ DTH, ELV, STP, OWC, MGL, Solar system**

The Contractor has both legal and contractual responsibility to formulate and implement SMART EHS Plan before proceeding for any construction activity. EHS Plan shall be prepared as per The Employer guidelines & shall include but not limited to – scope & type of activity like plumbing, firefighting, HVAC, ELV installation etc., & materials used for the activity, material storage arrangement, erection & installation, equipment's & machineries used etc. and shall be approved by The Employer Project In-Charge. The emergency management plan as per MBOCW (Rule 63), shall be part of the EHS Plan. The Contractor should submit EHS Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

35.8. **Elevators**

The Contractor has both legal and contractual responsibility to formulate and implement SMART EHS Plan before proceeding for any construction activity. EHS Plan shall be prepared as per The Employer guidelines and approval of The Employer Project In-Charge is mandatory. The emergency management plan as per MBOCW (Rule 63), shall be part of the EHS Plan. The Contractor should submit EHS Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

36. Preconstruction/ Site Establishment Safety

36.1 While planning for the trees cutting, Consider the size, species, health, and potential hazards (e.g., dead or diseased branches).

36.2 Strategically plan the direction of the tree's fall to avoid damaging property or endangering people.

36.3 While cutting the trees, use climbing belts, harnesses, and saddles properly. Inspect and maintain them before each shift of use. Make a drop zone marked with cones or other objects to avoid movement/working in an area where trimmings or tree removal may fall. Identify and avoid power lines and other energized objects.

36.4 To minimize the risk of snake bites during site cleaning, wear protective gear like gloves, long sleeves, and Gum boots/sturdy footwear. Be aware of surroundings, watch for snakes, and avoid putting hands in areas where you can't see.

36.5 Seek immediate medical attention for any snakebite, even if you're unsure if the snake was venomous.

36.6 During the surveying work, refer the details of ground conditions such as loose surfaces, potholes, sludge, and any other potentially life-threatening hazards, for the safety of survey personnel from unforeseen dangers. Be careful on uneven ground or around bushes and animals.

36.7 During the site surveying work, avoid working alone and have a supervisor nearby along with communication system and contact details for emergency.

36.8 Monitor the atmospheric conditions regularly and stop all survey, trees cutting etc., work immediately during storms, heavy rain, or strong winds to ensure the safety of all personnel

36.9 The contractor shall ensure that all dewatering pumps and temporary electrical installations are protected by 30mA RCCBs (Residual Current Circuit Breakers).

36.10 Only certified electricians shall perform the setup of dewatering pumps, electrical connections, and associated support systems.

36.11 For soil investigation, ensure only trained staff operate the drilling rigs. Drilling rigs along with associated lifting accessories shall be tested and certified by competent person and TPI Certificate shall be obtained prior commencing the activity.

36.12 Assess and mitigate potential hazards specific to the job site, such as weather conditions, underground utilities, and unstable ground.

36.13 Conduct regular inspections of the drilling rig and associated equipment to identify potential issues early and prevent equipment failure or related incidents.

36.14 Handle drilling fluids and other liquids with care to prevent spills and potential hazards.

36.15 To prevent fall-related incidents, all excavations, boreholes, and spoil materials must be adequately fenced off before vacating the worksite.

36.16 During the fencing work, all hot work at site like welding, gas cutting, brazing, heating etc shall be undertaken under Hot Work Permit & under supervision of contractor’s safety staff.

36.17 All hot work shall be backed by 6kg DCP/ABC Type portable extinguisher.

36.18 The Portable Fire Extinguishers shall be inspected once in a month for fitness conformity and logbook shall be maintained. The date of last inspection and maintenance shall be displayed prominently on fire extinguisher body.

36.19 The list of emergency telephone numbers including fire brigade number shall be prominently displayed at suitable locations (such as site office, gate security rooms, stores etc.) in bold distinct font.

36.20 Fire-fighting equipment should be used for fire-fighting purposes only, never place obstructions within 15 ft. of fire extinguishers.

37. Demolition Safety

37.1. On every demolition job, danger signs shall be conspicuously posted all around the structure and all doors and openings giving access to the structure shall be kept barricaded or manned except during the actual passage of workmen or equipment. However, provision shall be made for at least two independent exits for escape of workmen during any emergency.

37.2. During nights, red lights shall be placed on or about all the barricades.

37.3. The contractor shall ensure that no unauthorized person shall enter the site of demolition outside working hours; a watchman should be employed. In addition to watching, they shall also be responsible for maintaining all notices, lights and barricades.

37.4. All glass or similar material or article in exterior openings are removed before commencing any demolition work and all water, steam, electric, gas and other similar supply lines are put-off and suitably capped and the concerned department of the appropriate Government or local authority is informed and permission obtained wherever required before commencing such demolition work.

37.5. Wherever it is necessary to maintain water, gas or electric line or power during such demolition, such line shall be so located or protected with substantial coverings so as to protect it from damage and to afford safety to the building workers and the general public.

37.6. The contractor responsible for a demolition work shall examine the walls of all structures adjacent to the structure to be demolished to determine the thickness, method of support to such adjacent structures and in case, such adjacent structure is seems unsafe or may become unsafe during such demolition process, he shall not perform demolition activity until remedial measures like sheet piling, shoring, bracing, or similar other means so as to ensure safety and stability to such unsafe adjacent structure from collapsing are taken.

37.7. Any demolition of walls or partitions is proceeded in a systematic manner as per the standard safe operating practices and all work above each tier of any floor beams is completed before the safety of the supports of such beam is impaired.

37.8. Masonry is neither loosened nor permitted to fall in such masses or volume or weight as to endanger the structural stability of any floor or structural support.

37.9. No wall, chimney or other structure or part of a structure is left unguarded in such a condition that it may fall, collapse or weaken due to wind pressure or vibration.

37.10. In the case of demolition of exterior walls by hand, safe footing is provided for the building workers, in the form of sound flooring or scaffolds.

37.11. Walls or partitions which are to be demolished by hand are not left standing more than one storey high above the uppermost floor on which persons are working.

37.12. The contractor shall ensure safe access to and egress from every building is provided at all times in the course of demolition of building, by means of entrances, hallways, stairways or ladder runs which are so protected as to safeguard the building workers using such means from falling material or articles.

37.13. All steel structures are demolished column by column and tier by tier and every structural member which is being demolished is not under any stress and such structural member is suitably lashed/supported by boom cranes to prevent it from any uncontrolled swinging or dropping or falling.

37.14. large structural members are not thrown or dropped from the building but are carefully lowered by adopting suitable safe method/ mobile cranes etc.

37.15. Where a lifting appliance like a derrick is used for demolition, the floor on which such lifting appliance rests is completely planked over or supported, and such floor is of adequate strength to sustain bearing load for such lifting appliance and its operation.

37.16. Materials or articles are not stored or kept on platform, floor or stairways of a building being demolished.

37.17. An access to any stair ways or passageway is not affected or blocked by storing any material or article.

37.18. Suitable barricades are provided so as to prevent materials or articles from sliding or rebounding into any space used by the building workers.

37.19. Every opening used for the removal of debris from every floor which is not closed to access, except the top or working floor is provided with an enclosure from such floor to its ceiling, or such opening is so barricaded that no building worker has access to within a horizontal distance of six metres from such opening through which debris is being dropped.

37.20. The contractor shall ensure that a person responsible for demolition work makes continuous inspections during demolition process of such demolition work so as to detect any hazard resulting from weakened or deteriorated floors or walls or loosened materials or articles during such demolition process and that no building worker is permitted to work where such hazard exist unless remedial measures like shoring or bracing are taken to prevent such hazards.

37.21. Barricades and warning signs are erected along every side throughout the length and breadth of a building or other construction work to be demolished to prevent unauthorised persons from entering into the site of such building or other construction work during demolition operations.

37.22. During the demolition of an exterior masonry wall or a roof from a point more than twelve metres above the adjoining ground level of such wall or roof, if persons below such wall or roof are exposed to falling objects, suitable and safe catch platforms shall be provided and maintained at a level not more than six metres below the working level except where an exterior built-up scaffold is provided for safe and adequate protection of such persons.

37.23. Suitable and standard warning signs in accordance with national standards are displayed or erected at conspicuous places or position at the workplace.

37.24. Mechanical method of demolition: - The contractor shall ensure that the following requirements are fulfilled in case the mechanical method of demolition like use of swinging weight, clamshell bucket, power shovel, bulldozer or other similar mechanical methods are used for the purpose of demolition, namely: -

i. The building or structure or remaining portion thereof shall be not more than twenty-four metres in height.
ii. Where a swinging weight is used for demolition, a zone of such demolition having a radius of at least one and a half times the height of the structure or portion thereof being so demolished shall be maintained around the points of impact of such swinging weight.
iii. Where a clamshell bucket is being used for demolition, a zone of demolition shall be maintained within eight metres of the line of travel of such bucket.
iv. Where other mechanical methods are being used to affect total or partial collapse of a building or other construction work, there shall be maintained, in the area into which the affected portion of such building or other construction work may fall, a zone of demolition at least one and a half times the height of such affected portion thereof.
v. No person other than building workers or other persons essential to the operation of demolition work shall be permitted to enter a zone of demolition.
vi. Contractor shall abide by all the rules of MSBOCW Chapter XVI- Demolition & IS 4130:1991- Demolition of Buildings – Code of Safety

38. Excavation Safety

The contractor shall adhere to the following compliances while executing excavation activities at the construction site, including but not limited to:

38.1 Project Engineer- Appoint project engineer for safe operation of projects. Project engineer shall exercise overall control of operations and the activities

38.2 Responsible person- Contractor shall appoint responsible person with following duties and responsibilities.

i. Carry out smoothly such excavation
ii. Inspect and rectify any hazardous situation relating to such excavation
iii. Take remedial measures to avoid any unsafe practice or condition relating to the excavation
iv. Implement the EHS Plan of excavation

38.3 Warning Signs: Suitable warning signages or notices shall be displayed, required for the building worker carrying out the work.

38.4 Before excavation the contractor shall conduct total survey of the site for locating and removing or isolating existing utility services such as electric cables, high tension lines, water lines, PNG gas lines, sewage pipelines, chemical tanks etc and consult responsible government authority for necessary permission and control measure.

38.5 Where there is any doubt to the stability of structure adjoining the workplace or the area to be excavated, the project engineer shall arrange for measures like underpinning sheet piling, shoring, bracing or other similar means to support structure and prevent injury to any building worker working adjacent to the structure or damage to property or equipment adjacent to such structure.

38.6 Excavated soil and other construction materials shall be stacked at least 2m away from the excavation edges. Heavy equipment such as excavating machinery, and road traffic shall be kept back from the excavated sides at a distance not less than the depth of trench / excavated pit or at least 6mtr from the trench / excavated pit.

38.7 Vehicles such as trucks or wagons shall not be permitted to be driven near the tip of the excavated pit & care shall be exercise in locating roads leading to or from it.

38.8 Adequate & well anchored stop block shall be provided on the surface to prevent operating vehicles from falling accidentally into excavated pit.

38.9 Where any building worker engaged in excavation exposed with hazard of falling or sliding material or article from any bank or side of such excavation which is more than one and half meter above his footing, such worker is protected by piling and bracing against such bank side.

38.10 The excavation and its vicinity are checked by responsible person after every rain, storm or other occurrences carrying hazards and in case a hazard is noticed all such checking, adequate protection against slides and cave in to prevent such hazard is provided.

38.11 Temporary sheet piling installed for construction of retaining wall after excavation is not removed except on the advice of responsible person after any inspection carried out by responsible person.

38.12 Where banks of excavation are undercut, adequate shoring is provided to support the material or article over hanging such banks.

38.13 The contractor shall ensure the planks used for sheet piling in excavation is of sound material with adequate strength.

38.14 Shore and braces used in excavation are of adequate dimension and are so placed as to be effective for their intended purposes.

38.15 Earth supported shores or braces used in excavations bear against a footing of sufficient area and stability to prevent the shifting of such shores and braces

38.16 The contractor shall ensure safe access and egress to excavation where depth of excavation exceed one point five meters, by providing the ladders, staircase or ramp comply with the relevant national standards.

38.17 Where the depth of trench and excavation exceeds one point five meters suitable measures of falling shall be ensured such as adequately anchored MS Pipe railings. Whereas if the depth of trench and excavation exceeds four meters, improved protections in accordance with the design and drawing of professional engineer shall be ensured such as GI sheet fencing with adequate support and stability.

38.18 Any machinery used in excavation positioned and operated in such a way that such machinery does not endanger the operator of machinery or person in vicinity.

38.19 As per the nature of work and climatic conditions, contractor shall make the provision for adequate capacity and quantity of Dewatering pumps to handle the flooding conditions.

38.20 Contractor should abide by all the rules of MSBOCW Chapter XVII- Excavation and Tunnelling Works & IS 3764:1992 - Excavation work code of safety.

39. Structural Fabrication Work Safety

39.1. The contractor shall ensure the safety of workers employed for the erection or dismantling of steel structures and prefabricated structure is ensured free from danger by using appropriate means such as (i) ladders, gangways or fixed platforms; (ii) safety harness, lifelines, catch net or catch platform; (iii) power-operated mobile working platforms.

39.2. The work of erection or dismantling of structures work is carried out by trained & skilled workers under the competent supervision.

39.3. Steel or prefabricated structures should be designed and made that such structures can be safely transported or erected; and weight of each unit of such structures is clearly marked on such unit; ensure provision of railings with working platforms

39.4. Ensure the hooks and other devices built in or provided on the structural steel or prefabricated parts that are required for lifting and transporting such parts are examined P&M Incharge for any sign of damage before their use.

39.5. Lifting tools & tackles used for lifting of structural steel and prefabricated parts are provided with secure grip without damaging such parts and marked with the maximum permissible load.

39.6. Structural steel or prefabricated parts are lifted by such methods and appliances that prevent them from spinning accidentally, such as provision of tag line, guy rope etc.

39.7. Structural steel or prefabricated parts are provided with railings and working platforms before raising such parts to prevent any danger of falling of workers or materials.

39.8. Structural steel parts which are to be erected at a great height are, as far as practicable, assembled on the ground. When structural steel or prefabricated parts are being erected, a sufficiently extended area underneath the workplace shall be barricaded or guarded.

39.9. The erection gang should have adequate supply of bolts, washers, rivets, pins, etc, of the correct size. Enough number of bolts should be used in connecting each piece using a minimum of two bolts in a pattern to ensure that the joint will not fail due to dead load and erection loads.

39.10. The proper size, number and spacing of wire rope clamps should be used, depending on the diameter of the wire rope, they should be properly fixed.

39.11. The clamps should be checked as soon as the rope has been stressed, as the rope, especially if new, tends to stretch under the applied load, which in turn may cause it to shrink slightly in diameter. In addition, the clamps should be inspected frequently to be sure that they have not slipped and are tight enough.

39.12. Signalling while lifting shall be carried out through walkie talkie only no oral signal is allowed. Only trained signal man shall be deployed to perform signalling.

39.13. The safety nets being used shall be of fire-retardant quality as approved by the employer.

39.14. Structural fabrication work shall confirm to IS: 7205 – 1974 Safety Code for Erection of Structural steel work.

40. Waterproofing Work Safety

40.1. MSDS shall be made available and displayed at workplace for waterproofing chemicals and workers shall be made aware on health hazards & precautions to be taken.

40.2. Flammable chemicals shall be stacked separately & shall only have limited quantity of storage in workplace.

40.3. While handling hot air gun for jointing of sheets precaution to be taken to prevent contact with hot object like use of heat-resistant hand gloves.

40.4. Waterproofing sprayer, if used, shall be inspected prior to deployment at site & should check like condition of tubes, spray gun / pressure washer etc.,

40.5. While using waterproofing sprayers full body suit shall be used to prevent exposure to spray chemical getting absorbed by body.

40.6. All necessary PPEs like mask for respiratory protection, goggles for eye protection, gloves to prevent absorption of chemicals shall be used.

40.7. Roller brush used for the activity shall be of adequate length to reach most part while performing waterproofing & to avoid prevent height work.

40.8. Only trained & skilled workmen shall engage in handling paddle mixer, hot air gun & spray gun.

40.9. During the activity, avoid working alone & ensure supervisor along with communication system to contact in case of emergency.

40.10. Once the shift work is completed all the chemicals shall be stored back in chemical storage area & shall not be left at workplace.

41. Finishing Work Safety

41.1. Marble, granite etc., material shall be stacked in A-Frame within storage limit and secured with ropes/ratchet strap belt to prevent toppling of marble stacks.

41.2. Marble stacking & cutting shall be carried out at a safe distance to prevent inadvertent falling of marble on workmen.

41.3. Marbles stacked at floor level shall be secured to prevent fall.

41.4. Scrap marble pieces shall not be dumped at floor level; it shall be shifted to designated scrap area & shall be timely disposed.

41.5. Marble windowsill shall be precisely cut at floor itself for correct shape & size. No cutting activity shall be carried out at window which may lead to fall of material at below level.

41.6. All materials related to finishing activities like gypsum board, plasters, door frames, tiles, marbles etc., shall be stacked in orderly manner at designated storage areas provided to the contractor.

41.7. Chemicals, Flammable items, & other incompatible materials shall be stacked separately & away from materials storage area. Automated ceiling type fire extinguishers shall be provided in such chemical / flammable storage areas.

41.8. Aluminium mobile scaffolds shall be provided with adjustable platform, top & mid rail with toe-boards, heavy duty caster wheels with lock, access ladder etc with minimum loading capacity of 200 kg/m3 with required out-riggers for stability (to prevent toppling) shall be provided, inspected & tagged for its fitness by scaffold inspector.

41.9. Before commencement of work in service shafts, permit shall be obtained to remove or cut existing reinforcement fall protection mesh to ensure secondary fall protection in place to prevent exposure to fall hazard in shafts. While removing the said fall protection, lifeline with fall arrestor shall be used by the workmen.

41.10. While shifting materials from store to workplace, mechanical means shall be used to prevent from manual handling hazard wherever applicable.

41.11. While manually carrying & shifting materials, weight load shall be limited to 50kg for adult-man as prescribed by MBOCW rule 65. Heavy manual lifting is prohibited in the workplace.

41.12. Scissor lifts, MEWP, cherry picker used for finishing works shall be TPI certified by authorized competent person & shall be inspected daily before commencement of work.

41.13. Workmen engaged in such equipment's shall undergo vertigo test before deploying in such equipment's for height work.

41.14. Cranes used for lifting materials like marbles shall have all valid documents like RC, pollution certificate, TPI of crane for lifting & certificate of competent operator.

41.15. MSDS shall be made available for chemicals used for putty & plastering works and workers shall be made aware on health hazards such as inhalation, irritation, absorption & precautions like job rotation to reduce exposure, usage of respiratory protective equipment, full length clothing to prevent skin absorption etc., while handling such materials.

41.16. The contractor shall ensure fall restrain system tied in separate lifeline before deploying workmen at floor edge & at service shaft for internal painting & wall putty activity.

41.17. While carrying out painting activities, if hazardous painting is involved, the work location shall be restricted from workman other than painters to reduce exposure to chemical hazards. Job rotation of such painters shall be ensured to minimize handling hazardous paints.
41.18. Paints, varnishes, lacquers, thinners and other flammable materials shall be kept in a properly sealed or closed containers.
41.19. Use dedicated cabinets for flammable, corrosive, or toxic chemicals, grounded to prevent static build-up.
41.20. Never store paints with food, water, or drinking supplies.
41.21. No Smoking, No Hot Work, no open Flame signage shall be prominently displayed outside.
41.22. MSDS and immediate first aid measures shall be displayed.
41.23. Empty hazardous paint drums shall not be used as working platform & should be removed from work location to designated waste / scrap storage area and shall disposed through approved hazardous waste disposal vendors.
41.24. Buckets containing sand shall be kept ready for use in case of fire. Fire extinguishers, when required, shall be of foam type.

42. External Painting Safety

42.1. Rope suspended platform (RSP) shall have all mandatory safety devices like anti-tilt switch, upper stopper limit, antisway rope, emergency switch, safety device for locking during emergency etc., and these shall be examined by third party competent person before issuing TPI certificate. P&M department shall carry out daily inspection of said safety devices before starting of day activity.
42.2. If scaffold is used to perform external painting, clause 18 scaffold safety of this document shall be complied with.
42.3. In addition to mandatory inbuilt safety devices, separate fall grab arrestor & lifeline attached with the building structure shall be provided by the contractor to prevent inadvertent fall while working in RSP.
42.4. Daily inspection of RSP by technician & operator shall be ensured before commencement of activity.
42.5. Periodic maintenance of RSP shall be ensured as per OEM recommendations.
42.6. While relocating the RSP, authorized competent person shall verify & TPI certificate shall be obtained after every shifting & placement of RSP.
42.7. Apart from above, contractor should maintain one set of emergency rescue kit, especially for the rescue of RSP operator and other similar event/situation where person is suspended at height on safety harness due to sudden fall.
42.8. During adverse weather condition such as high winds, storm, lightening, heavy rain etc., work shall be suspended to prevent any untoward incident.
42.9. During such conditions RSP shall be lowered to ground level and shall not be left at hanging conditions to prevent fall or damages to structure.
42.10. While working at RSP, it is recommended that work shall be carried out during daytime only, & to be avoided during night hours.
42.11. Area underneath the painting activity shall be cordon off & precautionary signage shall be posted to prevent workmen entering such area.
42.12. Only skilled and trained workmen shall be engaged in external painting work in RSP with vertigo test before deploying for such activities.
42.13. Hand tools used while working in RSP shall be secured by lanyard to prevent inadvertent fall during the activity.
42.14. Area below the painting work should be barricaded and cleared from all sparks, open flame or any other ignition source

42.15. Daily safety instruction shall be carried out to workers carrying out parking bay marking including but not limited to - vehicle hazards, mandatory PPEs, chemical hazards & precautions, location of emergency point, first aid centre etc.,

42.16. Empty containers of hazardous paint shall be stacked separately and disposed through registered vendor

43. Façade Work Safety

43.1. All glass panels / stones / granite shall be transported & stored in A frame only. While stacking of these materials precautions to be taken to prevent inadvertent fall of material by providing ratchet belt to secure the materials.

43.2. All glass panels / stones / granite shall be stacked and stored on compacted & levelled surface, preferably on PCC floor.

43.3. The large size glass panel shall be preferably protected with film coating.

43.4. The contractor shall carefully plan for lifting and installations of façade, considering the weight of the glass panels, the height of the building, and potential wind conditions.

43.5. Operator of the Floor crane shall be competent and must be aware of the current and expected weather conditions before carrying out lifting activities.

43.6. Never lift glass panels with vacuum lifter during rain or when the panel is wet.

43.7. While using the vacuum lifter, examine the suction pads and seals for damage and wear.

43.8. Only skilled and trained riggers shall be engaged in façade installation activities with vertigo test before deploying for such activities.

43.9. All such activities shall be carried out under competent supervision.

43.10. All lifting equipment's, tools & tackles shall be examined by authorized competent person and a valid third-party certificate shall be obtained from him.

43.11. Check physical condition and test certificates of all lifting accessories like RSP, floor crane, and other lifting tools & tackles before commencement of activity.

43.12. Check that the safe working load (SWL) of the vacuum lifter is suitable for the load.

43.13. Whenever primary edge barricades need to be removed, ensure secondary protection like access restriction by lock & key or suitable barricades at floor level and prohibited signages shall be in place to prevent unauthorized entry into the panel installation areas to eliminate exposure to fall hazard.

43.14. The contractor shall ensure fall restrain system before deploying workmen at floor edge for glass panel installation activity.

43.15. Rope suspended platform (RSP) shall have all mandatory safety devices like anti-tilt switch, upper stopper limit, antisway rope, emergency switch, safety device for locking during emergency etc., and these shall be examined by third party competent person before issuing TPI certificate.

43.16. In addition to mandatory inbuilt safety devices, separate fall grab arrestor & lifeline attached with the building structure shall be provided by the contractor to prevent inadvertent fall while working in RSP.

43.17. Daily inspection of RSP by technician & operator shall be ensured before commencement of activity.

43.18. Periodic maintenance of RSP and safety devices shall be ensured as per OEM recommendations.

43.19. While relocating the RSP, authorized competent person shall verify & TPI certificate shall be obtained after every shifting & placement of RSP.

43.20. Apart from above, contractor should maintain one set of emergency rescue kit, especially for the rescue of RSP operator and other similar event/situation where person is suspended at height on safety harness due to sudden fall.

43.21. Area underneath the lifting activity shall be cordon off & precautionary signage shall be posted to prevent workmen entering such hazardous area.

43.22. After unwrapping of packed glass panels & other materials, waste shall be collected, stored in designated area & disposed in timely manner.

43.23. Broken / damaged glass panels shall be disposed immediately from workplace to designated scrap yard. While removing damaged glass panels special precautions to be taken to prevent falling of glass pieces and ground & below area should be barricaded with display of caution signage.

43.24. Glass panels shall be handled to workplace preferably by mechanical means to prevent manual handling hazard.

44. MEP- Electrical Work Safety

44.1. All materials related to MEP like cable trays, drums, conduits etc., shall be stacked in orderly manner, labelled & stacked away from flammable materials at designated storage areas provided to the contractor.

44.2. After completion of every shift, excess materials shall not be left at work floor, it should be taken back to material storage area.

44.3. Chemicals, Flammable items, & other incompatible materials shall be stacked separately & away from materials storage area. Automated ceiling type fire extinguishers shall be provided in such chemical / flammable storage areas.

44.4. Aluminium mobile scaffolds shall be provided with adjustable platform, top & mid rail with toe-boards, heavy duty caster wheels with lock, access ladder etc with minimum loading capacity of 200 kg/m3 with required out-riggers for stability (to prevent toppling) shall be provided, inspected & tagged for its fitness by scaffold inspector.

44.5. Before commencement of work in service shafts, permit shall be obtained to remove or cut existing fall protection (reinforcement mesh) & ensure secondary fall protection like lifeline with fall arrestor in place to prevent exposure to fall hazard in shafts.

44.6. While shifting materials from store to workplace, mechanical means shall be used to prevent from manual handling hazard wherever applicable.

44.7. While manually carrying & shifting materials, weight load shall be limited to 50kg for adult-man as prescribed by MBOCW rule 65. Heavy manual lifting is strictly prohibited in the workplace.

44.8. Workmen engaged in drilling, cutting, welding, activities shall be skilled & have prior experience to perform such activity and shall be periodically trained in usage of power tools & machineries.

44.9. Scissor lifts, MEWP, cherry picker used for MEP works shall be TPI certified by authorized competent person & shall be inspected daily before commencement of work.

44.10. Workmen engaged in such equipment's shall undergo vertigo test before deploying in such equipment's for height work.

44.11. Cranes used for such lifting shall have all valid documents like RC, pollution certificate, TPI of crane for lifting & certificate of competent operator.

44.12. Store incompatible chemicals separately (e.g., acids vs. bases, oxidizers vs. solvents).

44.13. No Smoking, No Hot Work, no open Flame signage shall be prominently displayed outside.

44.14. Workmen engaged in such height work shall undergo vertigo test before deploying.

44.15. Heavy cable drums shall be stacked at designated places & stopper block should be provided to prevent rolling of cable drums.

44.16. While rolling off the cable from drum, ensure cable drum jack is used or cable drum shall be upcycled.

44.17. Cable pulling shall be carried out by well-trained workmen, Essential tools include cable rollers, pulling wire, cable drum jacks, and measuring equipment shall be inspected prior to the activity.

44.18. Use cable rollers at sharp bends to prevent damage to the cable insulation. Avoid pulling cables around sharp corners without proper support

44.19. Contractor should follow the approved drawings for routing and securing cables in trays or ladders.

44.20. While installation & commissioning of substation only competent electricians shall be deployed & adequate supervision shall be provided. Electricians hand gloves (rated for voltage) shall be given while performing such duty. IS-4770:1991-rubber gloves.

45. MEP Work Safety

45.1. All materials related to MEP like pipes, ducts, fire alarms, chillers etc., shall be stacked in orderly manner, labelled & stacked away from flammable materials at designated storage areas provided to the contractor.

45.2. After shifting the pipes & other materials to workplace, it shall be stacked orderly, pipe chokes / stoppers should be provided to prevent inadvertent movement or rolling of pipes.

45.3. After completion of every shift, excess materials shall not be left at work floor, it should be taken back to material storage area.

45.4. Chemicals, Flammable items, & other incompatible materials shall be stacked separately & away from materials storage area. Automated ceiling type fire extinguishers shall be provided in such chemical / flammable storage areas.

45.5. Aluminium mobile scaffolds shall be provided with adjustable platform, top & mid rail with toe-boards, heavy duty caster wheels with lock, access ladder etc with minimum loading capacity of 200 kg/m3 with required out-riggers for stability (to prevent toppling) shall be provided, inspected & tagged for its fitness by scaffold inspector.

45.6. Before commencement of work in service shafts, permit shall be obtained to remove or cut existing fall protection (reinforcement mesh) & ensure secondary fall protection like lifeline with fall arrestor in place to prevent exposure to fall hazard in shafts.

45.7. While shifting materials from store to workplace, mechanical means shall be used to prevent from manual handling hazard wherever applicable.

45.8. While manually carrying & shifting materials, weight load shall be limited to 50kg for adult-man as prescribed by MBOCW rule 65. Heavy manual lifting is strictly prohibited in the workplace.

45.9. Workmen engaged in drilling, cutting, welding, activities shall be skilled & have prior experience to perform such activity and shall be periodically trained in usage of power tools & machineries.

45.10. Scissor lifts, MEWP, cherry picker used for MEP works shall be TPI certified by authorized competent person & shall be inspected daily before commencement of work.

45.11. Workmen engaged in such equipment's shall undergo vertigo test before deploying in such equipment's for height work.

45.12. Cables used to install at site like fire alarms cables, internet cables etc shall be properly rolled / stacked & secured to ensure it is not scattered at floor to prevent slip trip hazard.

45.13. Shifting & installation of heavy equipment's like chillers, cooling towers etc., shall be carried out as per approved method statement & risk assessment.

45.14. Cranes used for such lifting shall have all valid documents like RC, pollution certificate, TPI of crane for lifting & certificate of competent operator.

45.15. MSDS shall be made available for materials used for insulation of pipes & ducts and workers shall be made aware on health hazards such as inhalation, irritation, absorption & precautions like job rotation to reduce exposure, usage of respiratory protective equipment, full length clothing to prevent skin absorption etc., while handling such materials.

45.16. Paints, varnishes, lacquers, thinners and other flammable materials shall be kept in a properly sealed or closed containers.

45.17. Covered Storage area with PCC and sufficient slope towards collection pit. Provide bund of 110% of stored volume of fuel. Ensure adequate ventilation to prevent accumulation of flammable vapours

45.18. As far as possible, fuel drum shall be placed over drip tray

45.19. Store incompatible chemicals separately (e.g., acids vs. bases, oxidizers vs. solvents).

45.20. Use dedicated cabinets for flammable, corrosive, or toxic chemicals, grounded to prevent static build-up.

45.21. Never store paints with food, water, or drinking supplies.

45.22. No Smoking, No Hot Work, No open Flame signage shall be prominently displayed outside.

45.23. Painting & hot work of firefighting pipeline shall not be carried out at same locations to prevent fire hazards.
45.24. Empty hazardous paint drums shall not be used as working platform & should be removed from work location to separate waste / scrap storage area and shall disposed through approved hazardous waste disposal vendors.
45.25. During the hot work in MEP shafts of service pipelines, the bottom end of the pipes shall be closed by blind flange to prevent the spreading of hot spatters and subsequent risk of fire.
45.26. While installation & fixing of gas pipeline on outer face of the building g appropriate working platform like scaffold, RSP or any other approved mechanism with separate lifeline & fall arrestor shall be used.
45.27. Workmen engaged in such height work shall undergo vertigo test before deploying.

46. Elevator, Escalator and Travelators Work Safety

46.1. As far as possible all lift work inside the shafts shall be done using the falls-car system. The falls-car shall TPI tested and inspected daily before beginning day work. Use of independent lifelines with fall grab-arrester is mandatory
46.2. Safety facility removal permit: Removal of any type of safety protection facilities including barricades and lift shaft gates.
46.3. While removing the existing gates installed by Civil contractor for lift shaft protection, secondary fall protection measures (temporary lift shaft gates) shall be fabricated in advance and installed.
46.4. Scheme drawing shall be submitted prior erecting the scaffolding inside the lift shafts (confirming IS 3696-1 (1987): Safety code of scaffolds and ladders, Part 1)
46.5. Scaffoldings shall be erected under the guidance and inspected by competent scaffolding inspector.
46.6. A person working at height and exposed to risk of falling from height shall use full body safety harness with double lanyard & kinetic shock absorbers, fall grab-arrestor and lifelines ropes.
46.7. Only skilled, trained and experienced staff shall be deployed for work inside the shaft during elevator installation work.
46.8. Workmen engaged in elevator installation activity shall undergo Vertigo test.
46.9. Controlled access and egress shall be provided from the floors for the scaffolding inside the lift shaft, under lock and key.
46.10. During testing and commissioning of elevators, electrical Safety work permits & LOTO (lock out- tag out system) shall be used.
46.11. Use of bamboo scaffolds is prohibited in the project premises.

47. Infra Work Safety

47.1. During the installation of heavy Mild Steel (MS) gates, ensure that the gates are securely locked at their endpoints and that travel limits are set along the track to prevent any unintended movement or collapse.
47.2. During the canopy work, before commencing any heavy lifting work using lifting tools and tackles, a third-party inspection (TPI) must be conducted by a competent person, and a valid TPI certificate shall be made available at the site.
47.3. During work at height for canopy installation, a lifeline shall be installed along the periphery and designated fall zones, supported by rigid posts. All workers involved in height-related activities must wear full body safety harnesses, which shall remain securely anchored at all times.
47.4. During canopy work at height, safe access and egress shall be ensured through stair towers, ladders, scaffolding, or other suitable means, as appropriate to the specific task and site conditions
47.5. Traffic safety measures shall be ensured by diverting traffic or providing clearly visible signage, barricades, and traffic cones during pavement work, drainage work, stonework, or any other infrastructure development activities along operational roads.

47.6. All work at height shall be suspended during adverse weather conditions such as heavy winds, rain, lightning, storms, or any other unsafe environmental conditions.

47.7. Marble, granite etc., material shall be stacked in A-Frame within storage limit and secured with ropes/ratchet strap belt to prevent toppling of marble stacks.

47.8. Marble stacking & cutting shall be carried out at a safe distance to prevent inadvertent falling of marble on workmen.

47.9. Marbles stacked at floor level shall be secured to prevent fall.

47.10. Scrap marble pieces shall not be dumped at floor level; it shall be shifted to designated scrap area & shall be timely disposed.

47.11. Marble windowsill shall be precisely cut at floor itself for correct shape & size. No cutting activity shall be carried out at window which may lead to fall of material at below level.

47.12. All materials related to finishing activities like gypsum board, plasters, door frames, tiles, marbles etc., shall be stacked in orderly manner at designated storage areas provided to the contractor.

47.13. Chemicals, Flammable items, & other incompatible materials shall be stacked separately & away from materials storage area. Automated ceiling type fire extinguishers shall be provided in such chemical / flammable storage areas.

47.14. Aluminium mobile scaffolds used shall be provided with adjustable platform, top & mid rail with toe-boards, heavy duty caster wheels with lock, access ladder etc with minimum loading capacity of 200 kg/m3 with required out-riggers for stability (to prevent toppling) shall be provided, inspected & tagged for its fitness by scaffold inspector.

47.15. While shifting materials from store to workplace, mechanical means shall be used to prevent from manual handling hazard wherever applicable.

47.16. While manually carrying & shifting materials, weight load shall be limited to 50kg for adult-man as prescribed by MBOCW rule 65. Heavy manual lifting is strictly prohibited in the workplace.

47.17. Cranes used for lifting materials like marbles shall have all valid documents like RC, pollution certificate, TPI of crane for lifting & certificate of competent operator.

47.18. MSDS shall be made available for chemicals used for putty & plastering works and workers shall be made aware on health hazards such as inhalation, irritation, absorption & precautions like job rotation to reduce exposure, usage of respiratory protective equipment, full length clothing to prevent skin absorption etc., while handling such materials.

47.19. While carrying out painting activities, if hazardous painting is involved, the work location shall be restricted from workman other than painters to reduce exposure to chemical hazards. Job rotation of such painters shall be ensured to minimize handling hazardous paints.

47.20. Paints, varnishes, lacquers, thinners and other flammable materials shall be kept in a properly sealed or closed containers.

47.21. Use dedicated cabinets for flammable, corrosive, or toxic chemicals, grounded to prevent static build-up.

47.22. Never store paints with food, water, or drinking supplies.

47.23. No Smoking, No Hot Work, no open Flame signage shall be prominently displayed outside.

47.24. MSDS and immediate first aid measures shall be displayed.

47.25. During the deep cleaning from external face of building, the Contractor shall mandatorily ensure that the independent lifeline of 12mm/ 14mm Polyamide rope of approved quality and fall-grab arrester is provided. Ensure that the lifelines are independently fixed to the building structure or any other rigid support and not tied with the equipment/ platform on which the person is working.

47.26. Workmen engaged is deep cleaning activity on external face of building shall undergo Vertigo test.

47.27. Flammable chemicals shall be stacked separately & shall only have limited quantity of storage in workplace.

47.28. During the activity, avoid working alone & ensure supervisor along with communication system to contact in case of emergency.

47.29. Once the shift work is completed all the chemicals shall be stored back in chemical storage area & shall not be left at workplace.

48. Acknowledgement by Contractor

ACKNOWLEDGEMENT OF SAFETY RULES AND REGULATIONS BY CONTRACTOR

I, _____ (Name) _____(Designation) of _____(The Contractor) being a Contractor for the project at _____, do hereby confirm that I have been briefed by the Contractor Project In-Charge and Safety Manager on Safety Rules & Regulations. I fully understand the requirements and agree to abide by the same while working at project and shall ensure safety of employees. Any contravention of the rules and regulations and incident at project on our part shall render us fully accountable to all the consequences.

I confirm that I am a Director/ Authorised and appointed representative of the Contractor.

My contract value for this project is Rs. _____Crores / Lakhs. The Contract period is from _____to _____.

_____	_____
Signed by:	Witnessed by:
Employer/ Project Manager.	Contractor / Project In-Charge
Name: _____	Name: _____
Designation: _____	Designation: _____
Contact No: _____	Contact No: _____

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04th December 2025

Index (Table B): Safety, Health and Environmental Rules and Regulations, **R06 01st August 2025**

Sr. No.	Package No.	A	B	C	D	E	F	G	H	I	J	K	L	M	Page No. wrt Annexure 03, Index (Table A)
	Master Package	Civil	Civil	Civil	Civil	Civil	Civil	Finishing	Façade	Façade	MEP Electrical	MEP Mechanical	MEP	Infra	
	Sub Package	Concrete Shuttering Rebar	site cleaning, site fencing, site enabling works, soil investigation, site survey works	Demolition of existing structure	Earthwork, Pile and rock anchor work, soil retention	Heavy Fabrications (Design and Build)	Waterproofing	Integrated Finishing, Internal and External Doors	External Painting, Parking Bay Marking	Flat Façade, Glass Railing, GRC/ GFRC, Other Façade	Internal Electrical works, Substation	Plumbing water supply and drainage works, infra plumbing, firefighting, HVAC, IBS/ FTTH/ DTH, ELV, STP, OWC, MGL, Solar system	Elevators	Hardscape, Softscape, Main Gate, Sales & Show Flat (Civil), Sales & Show flat (ID & mep), Infra civil works, other Misc. support services	
1	Introduction	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	4
2	Applicability of Rules and Regulations	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	4
3	General Safety and Health Rules	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	4
4	Method Statement and Hazard Identification & Risk Assessment	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	6
5	Safety Staff	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	6
6	Personal Protective Equipment	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	8
7	Fire Prevention and Protection	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
8	Housekeeping and Hygiene	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	10
9	Lifting and Hoisting Machinery	✓	✓	✓	✓	✓		✓		✓	✓	✓	✓	✓	12
10	Compressed Gas Cylinders	✓	✓	✓	✓	✓				✓	✓	✓	✓	✓	14
11	Electric Arc Welding & Gas Cutting (Hot Work)	✓	✓	✓	✓	✓				✓	✓	✓	✓	✓	15
12	Electrical Safety	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	16
13	Portable Electrical Hand Tools and Equipment's	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	18
14	Hand Tools Safety	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	18
15	Machine and Hand Tools Guarding	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	18
16	Fall Prevention and Protection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	19
17	Lifelines & Fall-grab Arresters	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		25
18	Working Platforms and Ladders	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	25
19	Scaffolding Safety	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	25
20	Incident Reporting and Investigation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	26
21	Safety Trainings	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	26
22	Safety Signages	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	27
23	Welfare of Building Workers	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	27
24	Health Management and Medical Facility	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	29
25	Passenger Hoist and Material Hoists	✓				✓									30
26	Work Permit	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	30
27	Legal Compliance and Register	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	31
28	Penalty Against Safety Violations and Accident/ Incident	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	32
29	Additional Safety Rules and Regulations	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	35
30	Abbreviations	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	35
31	Confined Space Safety	✓				✓	✓				✓	✓		✓	35
32	Diesel Storage and Flammable Material	✓		✓	✓	✓									36
33	Ambient Air and Noise Pollution Monitoring	✓		✓	✓	✓									36
34	Local Governing body statutory requirement for Air Pollution Mitigation	✓		✓	✓	✓		✓							36
35	Environment Health and Safety Plan (EHS Plan)	✓		✓	✓	✓				✓	✓	✓	✓		38
36	Preconstruction/ Site Establishment Safety		✓												40
37	Demolition Safety			✓											41
38	Excavation Safety				✓										42
39	Structural Fabrication Work Safety					✓									44
40	Waterproofing Work Safety						✓								44
41	Finishing Work Safety							✓							45
42	External Painting Safety								✓						46
43	Façade Work Safety									✓					47
44	MEP – Electrical Work Safety										✓				48
45	MEP Work Safety											✓			49
46	Elevator, Escalator and Travelators Work Safety												✓		50
47	Infra Work Safety													✓	50
48	Acknowledgment by Contractor	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	52
	Total Count	**36**	**29**	**35**	**34**	**37**	**29**	**30**	**28**	**32**	**33**	**33**	**32**	**31**	

Index (Table B): Safety, Health and Environmental Rules and Regulations, **R06 01st August 2025**

Sr. No.	Package No.	A	B	C	D	E	F	G	H	I	J	K	L	M
	Master Package	Civil	Civil	Civil	Civil	Civil	Civil	Finishing	Façade	Façade	MEP Electrical	MEP Mechanical	MEP	Infra
	Sub Package	Concrete Shuttering Rebar	site cleaning, site fencing, site enabling works, soil investigation, site survey works	Demolition of existing structure	Earthwork, Pile and rock anchor work, soil retention	Heavy Fabrications (Design and Build)	Waterproofing	Integrated Finishing, Internal and External Doors	External Painting, Parking Bay Marking	Flat Façade, Glass Railing, GRC/ GFRC, Other Façade	Internal Electrical works, Substation	Plumbing water supply and drainage works, infra plumbing, firefighting, HVAC, IBS/ FTTH/ DTH, ELV, STP, OWC, MGL, Solar system	Elevators	Hardscape, Softscape, Main Gate, Sales & Show Flat (Civil), Sales & Show flat (ID & mep), Infra civil works, other Misc. support services
4	Method Statement, Hazard Identification & Risk Assessment and Safety Audit													
	5.1. Method Statement and HIRA	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	5.2. Safety Inspections	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	5.3. Safety Audit	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable
	5.4. Control Blasting (The Explosives Rules, 2008	Not Applicable	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable
5	Safety Staff													
	DGM/ Sr. Safety Manager (DGM in case of more than 3 Towers)	✓	Not Applicable	Not Applicable	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	✓	Not Applicable	Not Applicable	✓	Not Applicable
	Safety officer	✓	Not Applicable	✓	✓	✓	Not Applicable	✓	✓	Not Applicable	✓	✓	Not Applicable	Not Applicable
	Safety Supervisor	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
9	Lifting and Hoisting Machinery	Applicable upto 9.10.	Applicable upto 9.10.	Applicable upto 9.10.	All sub clauses applicable	Applicable upto 9.10.	Not Applicable	Applicable upto 9.10.	Not Applicable	Applicable upto 9.10.	Applicable upto 9.10.	Applicable upto 9.10.	Applicable upto 9.10.	Applicable upto 9.09.
10	Compressed Gas Cylinders	upto 10.18	upto 10.18	upto 10.17, 10.19	upto 10.17	upto 10.17	Not Applicable	Not Applicable	Not Applicable	upto 10.18	upto 10.17	upto 10.17	upto 10.17	upto 10.17
11	Electric Arc Welding and Gas Cutting (Hot work)	✓	All except 11.14	✓	✓	✓	Not Applicable	Not Applicable	Not Applicable	✓	✓	✓	✓	✓
12	Electrical Safety	✓	upto 12.38	✓	✓	All except 12.7	All except 12.3, 12.4, 12.45, 12.46	✓	All except 12.3, 12.4, 12.21, 12.44, 12.45, 12.46	All except 12.7	✓	✓	✓	✓
16	Fall Prevention and Protection	Applicable till 16.6	Not Applicable from 16.1 till 16.9	Only 16.1.i., 16.1.iv. And 16.1.v. applicable	Only 16.1.iii., 16.1.iv. And 16.1.v. applicable	All applicable except 16.2.i., 16.2.ii., 16.7, 16.8 and 16.9	Only 16.7 and 16.8 applicable	Only 16.7, 16.8 and 16.9 applicable	Not Applicable from 16.1 till 16.9	Only 16.8 applicable	Only 16.7 and 16.8 applicable	Only 16.7 and 16.8 applicable	Only 16.1.iv. 16.1.v. 16.1.vii. Applicable	Only 16.1.iv. And 16.1.vi. Applicable
17	Lifelines & Fall-grab Arresters	✓	Not Applicable	Only 17.1 Applicable	Only 17.1 Applicable	Only 17.1 Applicable	Only 17.1 Applicable	Only 17.1 Applicable	✓	✓	Only 17.1 Applicable	Only 17.1 Applicable	Only 17.1 Applicable	Not Applicable
21	Safety Trainings													
	21.1. Safety Induction of Employees	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	21.2. Work Related Trainings	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	21.3. Toolbox Talk Training	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	21.4. Safety Awareness Campaign	✓	Not Applicable	✓	✓	✓	Not Applicable	✓	Not Applicable	✓	✓	✓	✓	Not Applicable
22	Safety Signages	✓	✓	✓	✓	✓	upto 22.3.	✓	upto 22.3.	✓	✓	✓	✓	✓
23	Welfare of Building Workers													
	23.1. Drinking Water	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	23.2. Crèche	✓	Not Applicable	Not Applicable	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	✓
	23.3. Canteen	✓	Not Applicable	Not Applicable	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	✓
	23.4. Accommodation	✓	Not Applicable	✓	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	✓
	23.5. Shelter	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	23.6. Labour Toilet, Urinals and Pest Control	✓	Not Applicable	✓	✓	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable
24.0	Health Management and Medical Facility													
	24.1. First Aid Centre	✓	Not Applicable	✓	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable
	24.2. Preliminary Medical Examination of Workers	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	24.3. Medical Examination of Operators and Drivers	✓	✓	✓	✓	✓	Not Applicable	✓	✓	✓	✓	✓	✓	✓
	24.4. Medical Examination Record	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	24.5. Hospital Tie-Up for Medical Emergency	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	24.6. Ambulance Van	✓	Not Applicable	✓	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	✓	Not Applicable
	24.7. Medical Officer	✓	Not Applicable	✓	Not Applicable	✓	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable
	24.8. Stretcher	✓	✓	✓	Not Applicable	✓	✓	✓	✓	✓	✓	✓	✓	Not Applicable
	24.9. First Aid Box	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
32	Diesel Storage and Flammable Material	Applicable upto 32.9	Not Applicable	Applicable upto 32.9	Applicable upto 32.9	All except 32.7, 32.8	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable	Not Applicable
35	Environment, Health and Safety Plan of Project (EHS Plan)	✓	Not Applicable	✓	✓	✓	Not Applicable	Not Applicable	Not Applicable	✓	✓	✓	✓	Not Applicable